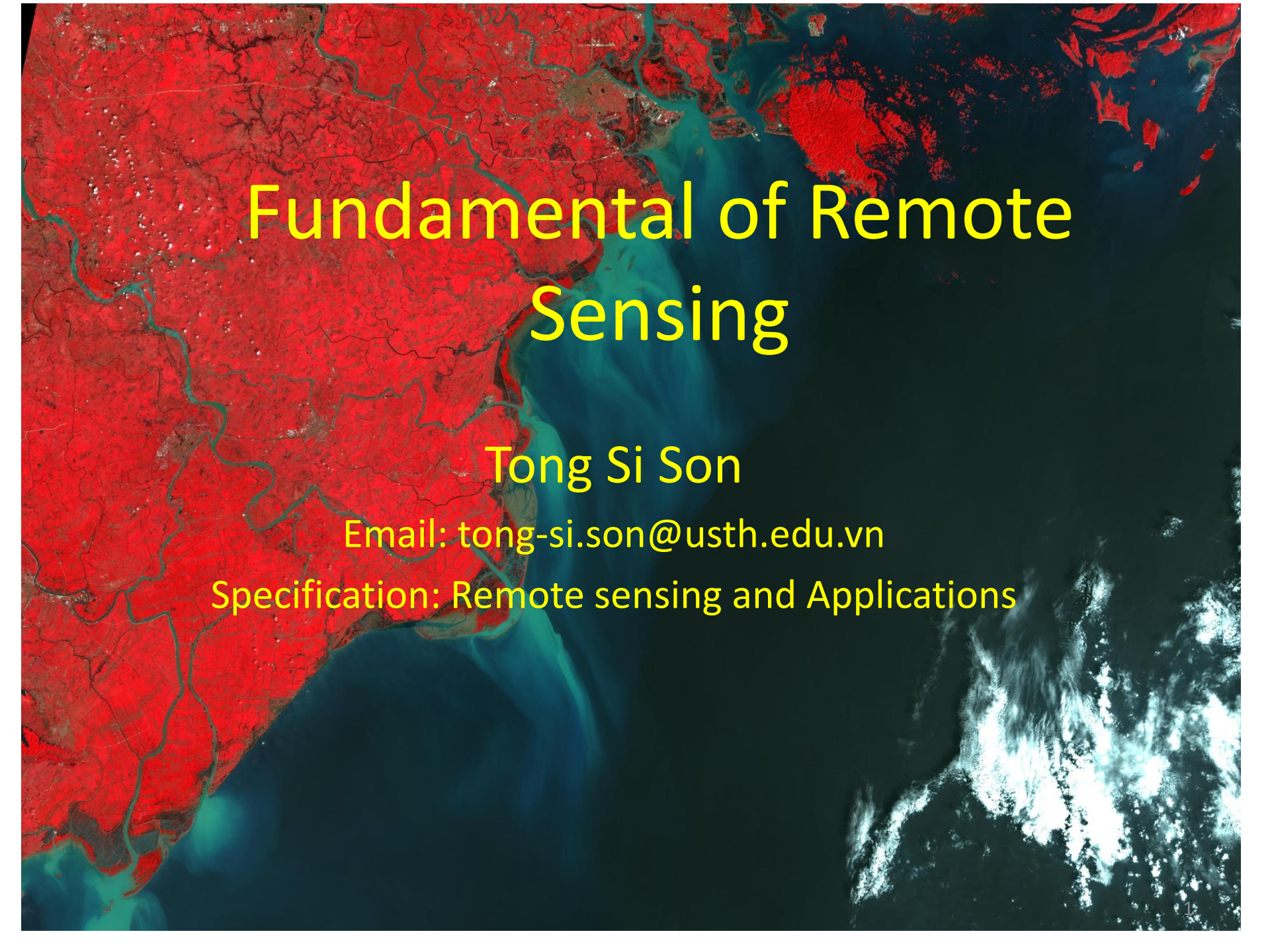


A satellite image of a coastal region. The land is colored red, showing a complex network of rivers and waterways. The water is dark blue. In the bottom right corner, there is a bright, white, cloud-like feature. The text is overlaid on the image.

Fundamental of Remote Sensing

Tong Si Son

Email: tong-si.son@usth.edu.vn

Specification: Remote sensing and Applications

Structure Of The Course

- **Time**

36 hours: Lecturer: 21 h, Practice: 15 h

- **Purposes**

Introduce the principles, characteristics, applications of remote sensing to students who are beginners in this field.

- **Contents**

1. **Overview of Remote sensing**

2. Physical basis of Remote sensing

3. Remote sensing system

4. Satellite image characteristics

5. Satellite image pre-processing

6. Extract information from satellite data and accuracy assessment

- **References**

https://webapps.itc.utwente.nl/librarywww/papers_2009/general/principlesremotesensing.pdf

https://www.nrcan.gc.ca/sites/www.nrcan.gc.ca/files/earthsciences/pdf/resource/tutor/fundam/pdf/fundamentals_e.pdf

Lillesand et al.. 2008, Remote sensing and Image interpretation, Wiley&Sons

George Joseph, 2015, Fundamentals of Remote sensing, Universities Press (India) Private Limited

[John Jensen](#), 2006, Remote Sensing of the Environment: An Earth Resource Perspective 2nd Edition, Pearson publisher

Section 1. Overview of Remote Sensing

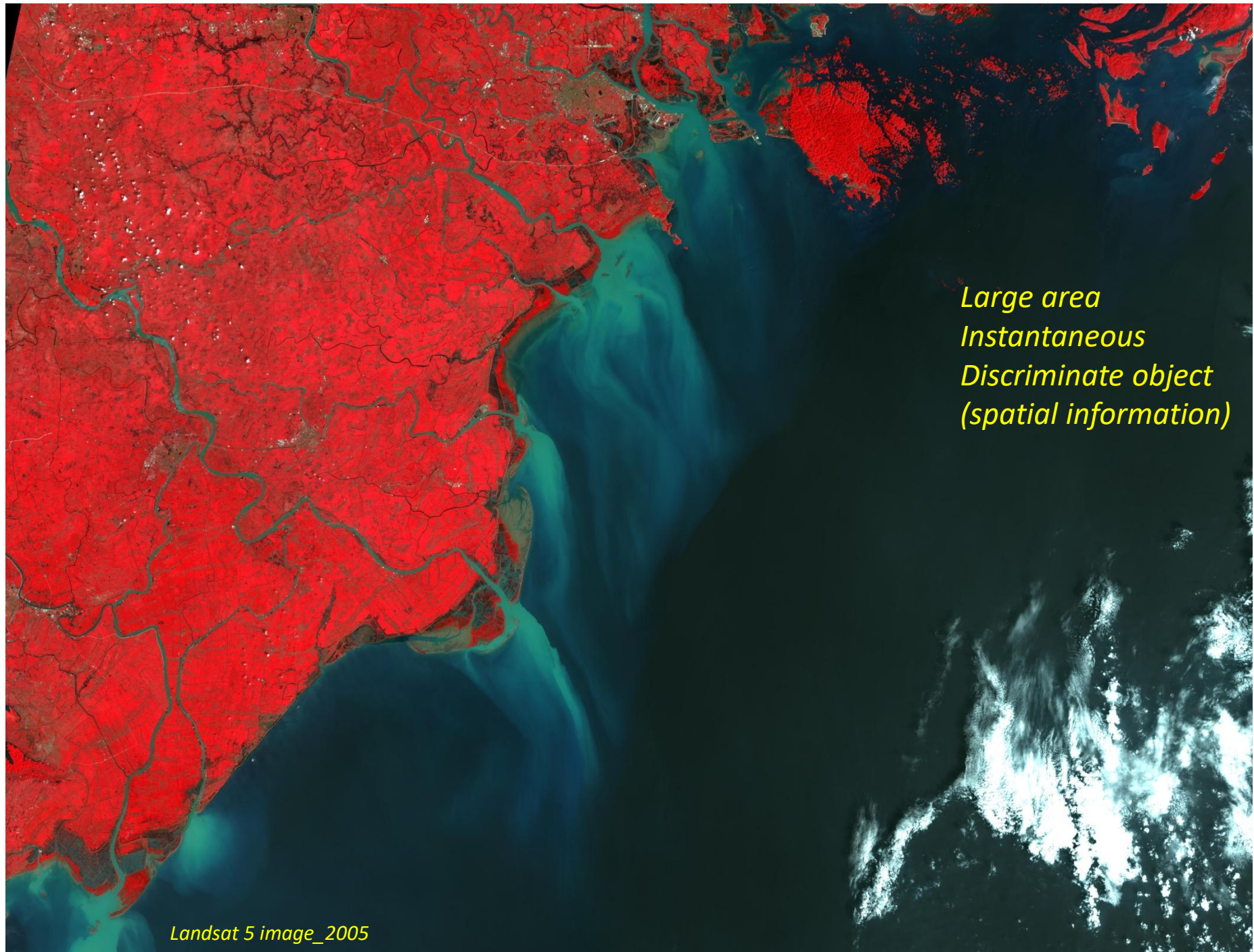
1.1 Definition of Remote sensing

1.2 History of Remote sensing

1.3 Physical basics of Remote sensing

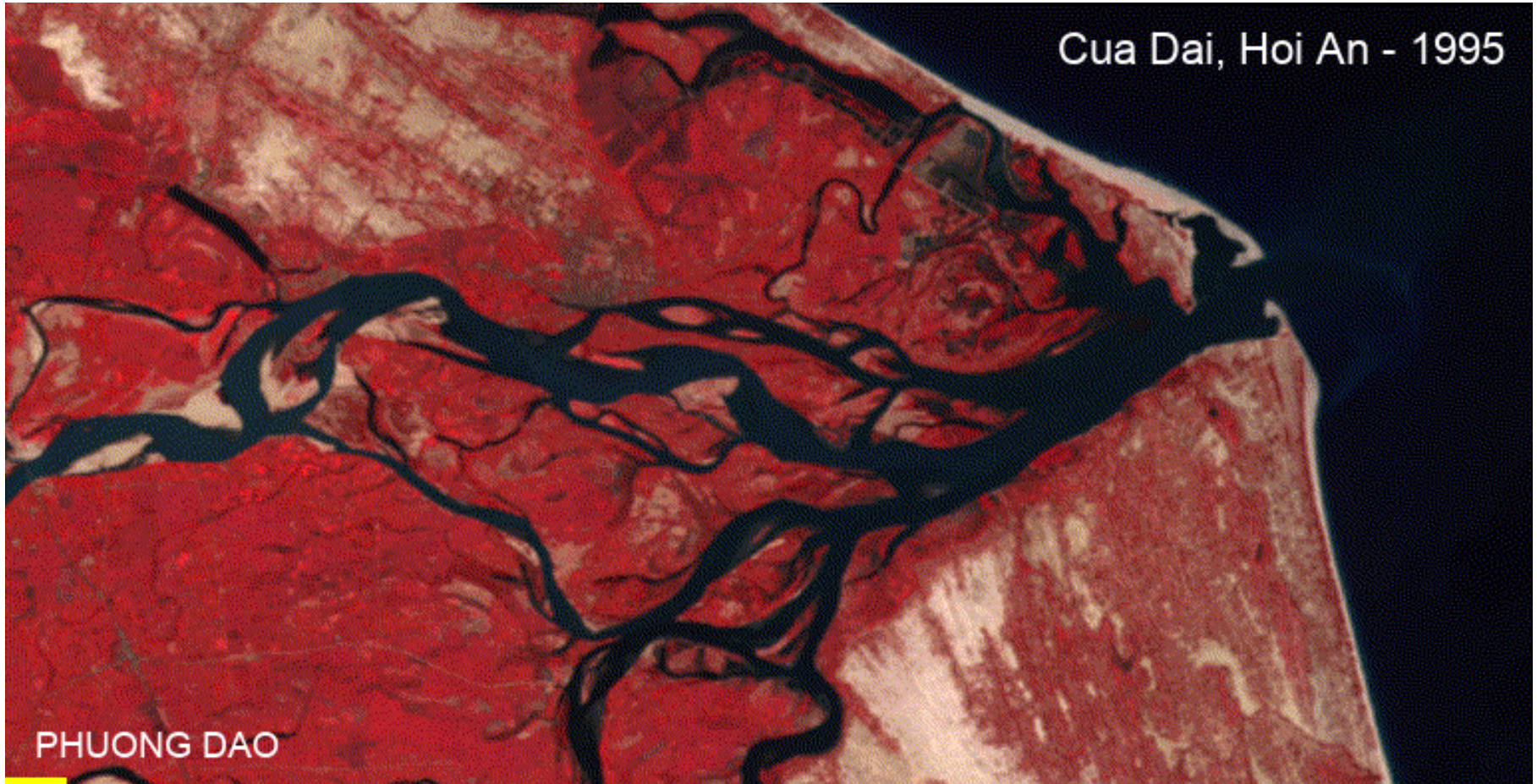
1.4 Remote sensing systems

1.5 Application of Remote Sensing



*Large area
Instantaneous
Discriminate object
(spatial information)*

Landsat 5 image_2005

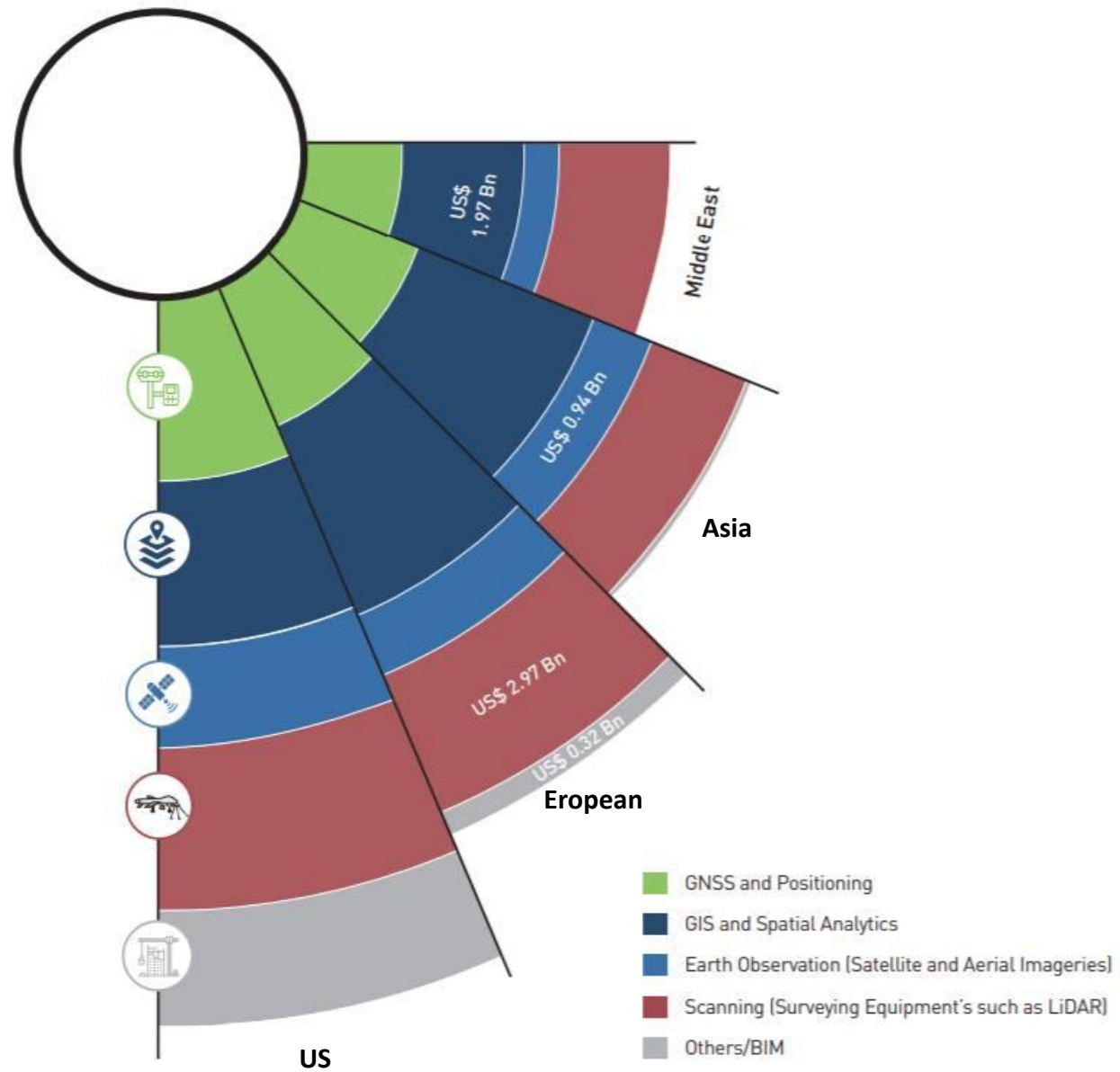


<https://www.facebook.com/daoducphuong.tdb>

- Time variation (temporal variation)
- Importance for planning (decision on build or not build a resorts)
- Land, soil, environment management



Geospatial Market in AEC Industry, by technology segment, 2019 (US\$ Billion)



1.1 Definition of Remote sensing



How do you recognize the differences between two moneys?



Ho Chi Minh $\neq$ Franklin
500.000 $\neq$ 100
Smaller $\neq$ Bigger
Green $\neq$ Black

Reflectance of light,
spectrum, electromagnetic
waves to eyes, films, sensors

Definition of remote sensing

The Experts say "Remote Sensing is..."

...**techniques** for **collecting image** or **other forms of data** about an object from measurements made at **a distance from the object**, and the processing and analysis of the data (RESORS).

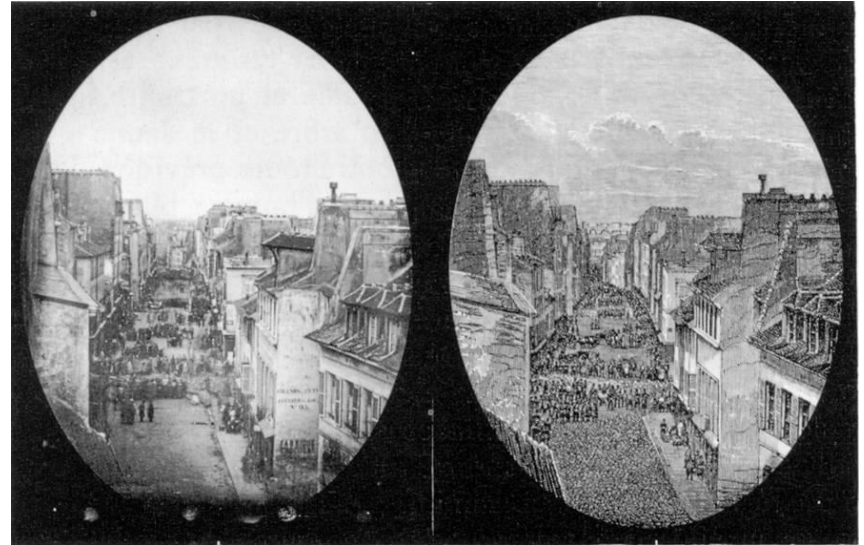
"...the **science** (and to some extent, art) of acquiring information about the Earth's surface **without actually being in contact with it**. This is done by sensing and **recording reflected or emitted energy** and processing, analyzing, and applying that information." (CCRS)

1.2 History of Remote sensing

(material by Jensen, 2000)

Remote sensing is at the crossroads of three major inventions:

1. Photography
2. Aeronautics
3. Propulsion



Barricade à Saint-Maur Popincourt, février 1848; daguerréotype et bois gravé paru dans « L'Illustration ».



Development of Photography

Basis as early as 1812. First photograph in 1827 by Nicéphore Niepce using a polished pewter plate covered with a petroleum derivative, the so-called bitumen of Judaea. Daguerre continued the work after Niepce died.

The First Photograph



- The first photograph was obtained by Joseph Nicéphore Niepce of his French estate courtyard in 1827.
- The exposure lasted 8 hours and used an emulsion of Bitumen of Judea (a kind of asphalt).

The viewpoint of Gras at Saint-Loup-de-Varennnes in 1827

Development of Photography

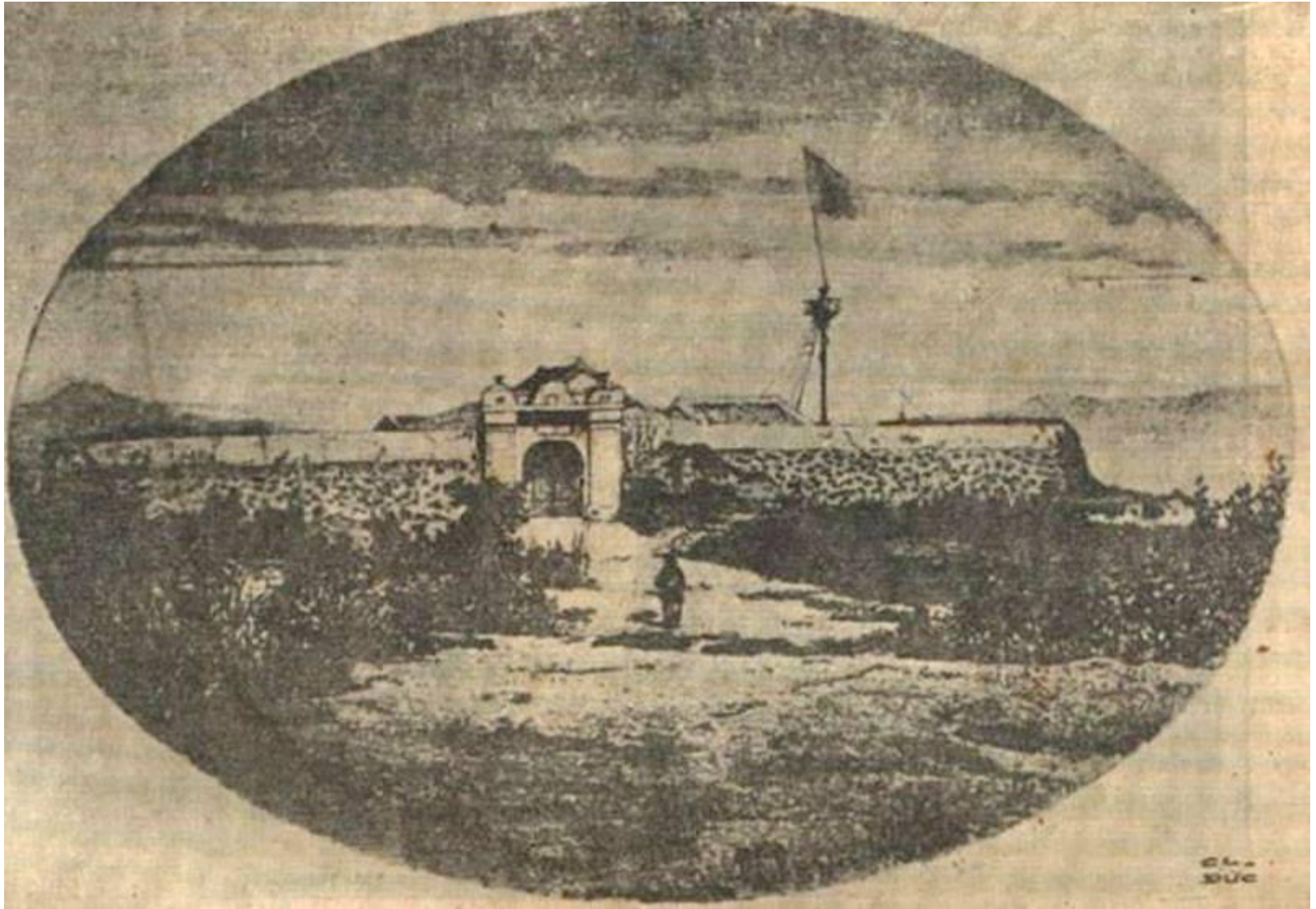


Camera made by Maison Susse Frères in 1839,
with a lens by Charles Chevalier



Le boulevard du Temple, Paris, 1838

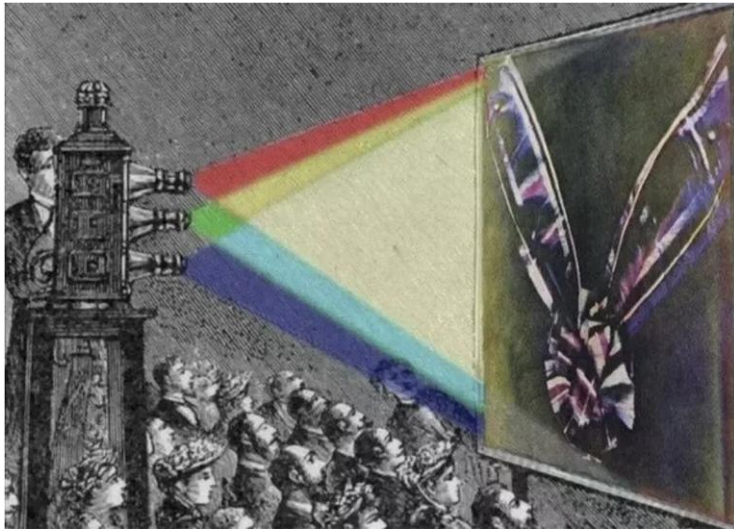
Development of Photography



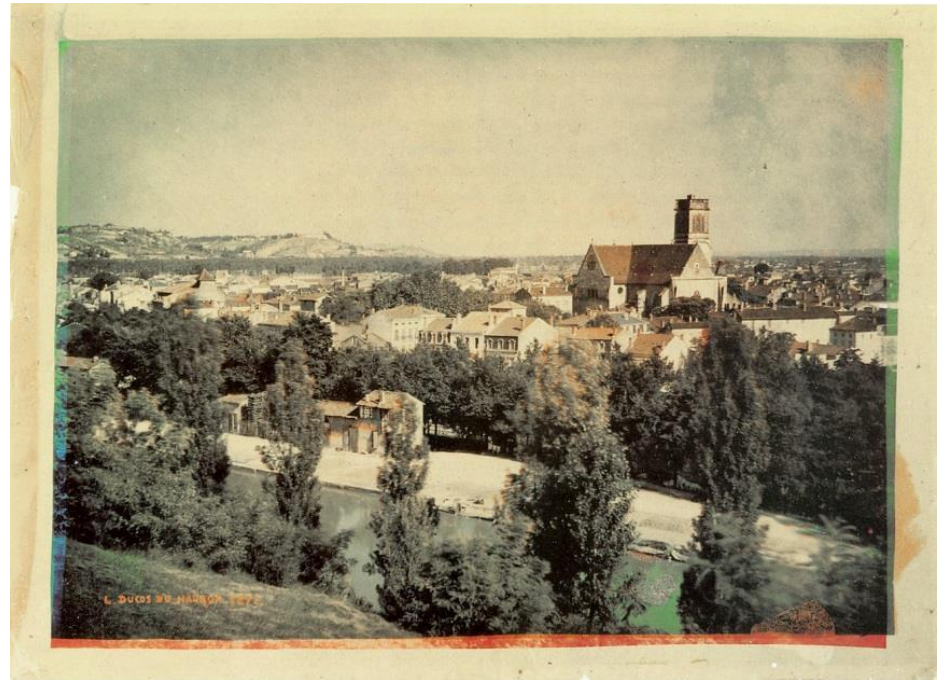
The first image captured Vietnam, in Đồn Hai, Đà Nẵng by Mr. Jules Itier on **31rd May, 1845** using Daguerre's camera made in 1839

Development of Photography

- 1855 : **J.C. Maxwell** developed the theory of additive synthesis of color using 3 basis color Red, Green, Blue.
- 1869 : **Charles Cros** and **Louis Ducos du Hauron** presented the indirect color photography principle (soustractive method).



Thomas Sutton (1861) took the a colored ribbon using the method of Maxwell developed early in 1855



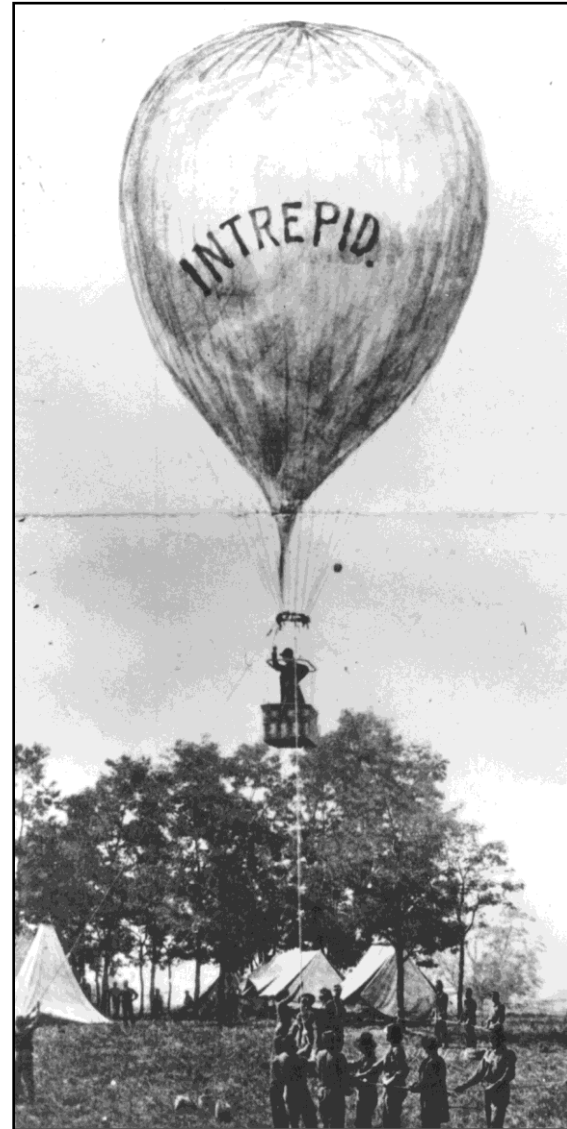
Colored photo of Agen city captured by Louis Ducos du Hauron (1877)

Camera Attached on Balloon

Balloon *Intrepid*

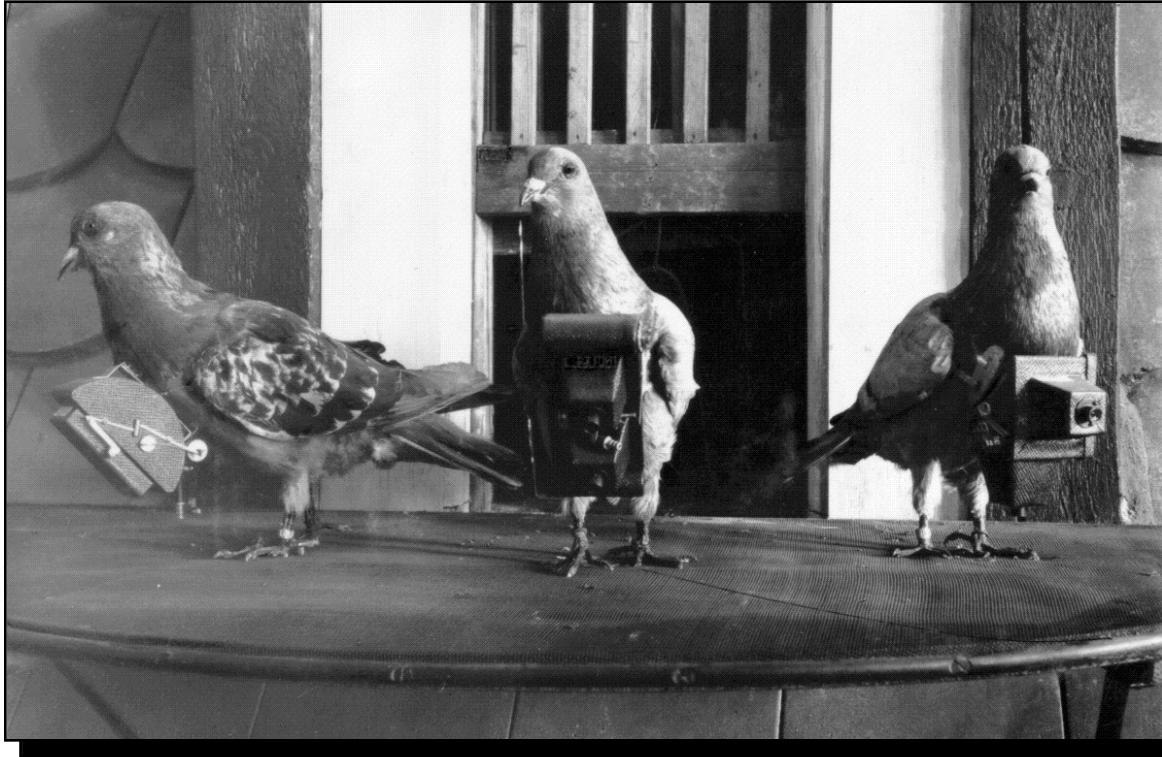


- Oblique aerial photograph of downtown Boston obtained by Samuel A. King and J. W. Black from a balloon at an altitude of 1,200 ft. on October 13, 1860.
- First aerial photograph taken from a captive balloon in the United States.



- *Intrepid* tethered during the Civil War battle of Fair Oaks on June 1, 1862 for a **reconnaissance**. (High speed of exposition)

Camera Attached on Pigeons



- In 1903, Julius Neubronner patented a breast-mounted camera for carrier pigeons that weighed only 70 grams.
- A squadron of pigeons is equipped with light-weight 70-mm aerial cameras.



- Oblique aerial photograph of a European castle obtained from a camera mounted on a carrier pigeon. The pigeon's wings are visible.

Aeronautics and Photography

The airplane has been invented in 1903 by the Wrights.

30 October 1908 – flight by **Farman** (30km) between Bouy and Reims.

The first aerial motion picture in 1909 in Centocelli (Italy)



Flight by Thoman in 1911, Issy-les-Moulineaux, crash when landing

First World War :

Air reconnaissance:

Mapping of threatened operations, use of toxic gases



Carnoy et Montauban de Picardie

British photography: June 1916

Aeronautics and Photography

Curtiss AH-13



- Pilot and aerial photographer with a Graflex aerial reconnaissance camera in 1915.

Aeronautics and Photography



One of the first Aerial Photos in Vietnam in 1910

Aeronautics and Photography



Bach Mai Airport in 1924

Aeronautics and Photography

Boeing B-29 and Photogrammetric Equipment



- Getting ready to obtain aerial photography of a nuclear weapons test at Bikini Atoll on July 25, 1946.

Propulsion

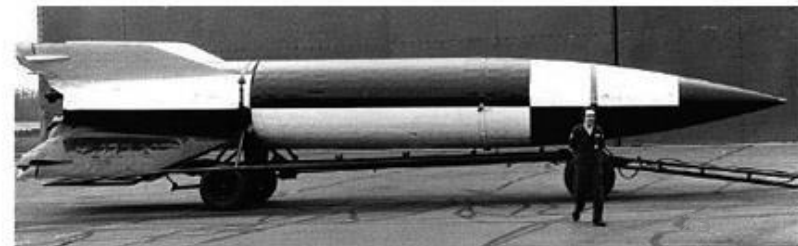
Second World War: Bombers and Fighters

The rockets are improved by the Germans (V-1 then V-2).

In 1942 a German rocket reached 90km of altitude. Opened the era of space observation



First spatial image taken on October 24, 1946
over New Mexico (from a V2). Altitude: 104 km



Propulsion



First satellite : Sputnik 1
(USSR) in 1957

First man in orbit: Gagarine
in 1961

First man in the space:
Leonov in 1965

The first satellite Sputnik 04th October 1957 by USSR

Propulsion

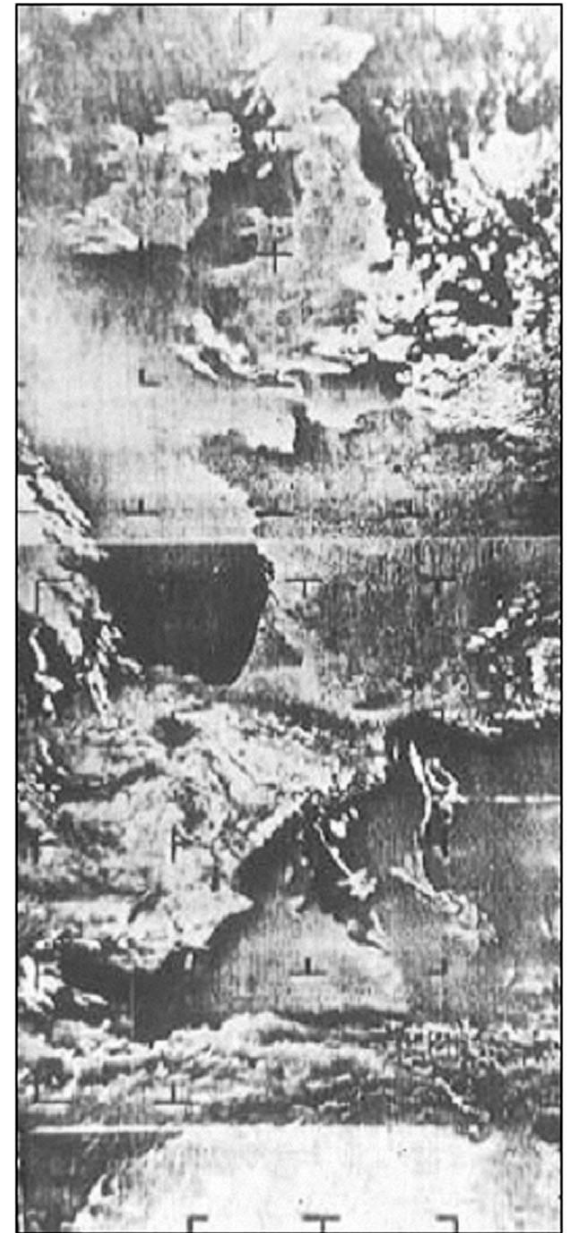
First satellites are mainly used for meteorology

TIROS 1-10 (1960-1965)
+ TIROS M (1970), TIROS N (1978)

NOAA 1-19 (1960-2009)
*NOAA 20 planned in 2017 (Joint Polar
Satellite System-1)*

NIMBUS 1-7 (1964-1978)

ESSA 1-9 (1966-1969)



Tiros 1, 1960, First space view of France

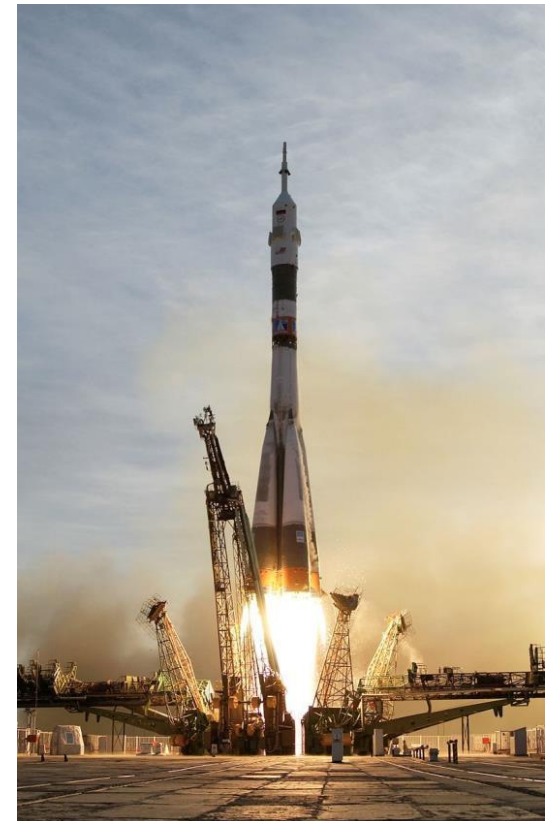
Propulsion

Ariane 5 the launchers for sending the largest payload (satellite):

- 20 tons in low orbit
- 10 tons in geostationary orbit

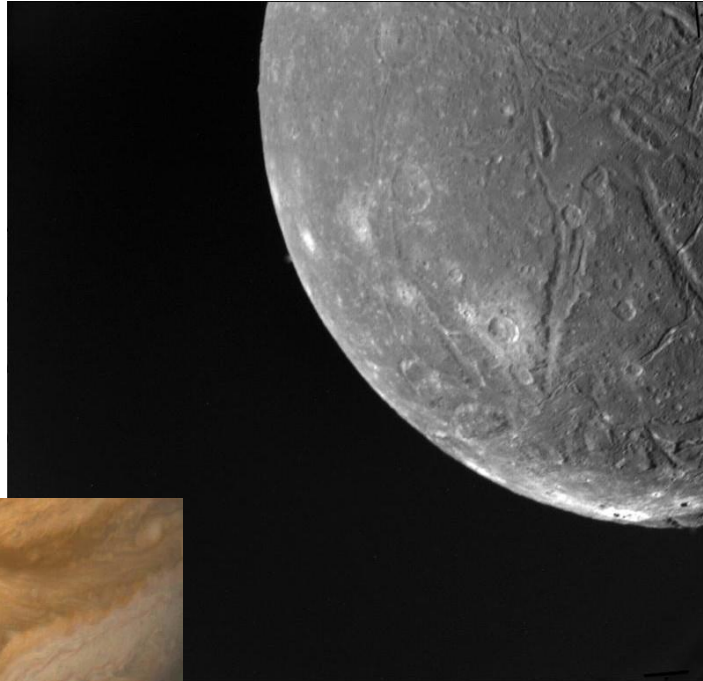


Ariane 5, 12 February 2005



Soyuz 5, 14 October 2004

Propulsion



*Photo of Ariel, a
satellite of Uranus
captured by Voyager 2
(January 1986)*



*Photo of Pluton captured by New
Horizons (13 July 2015)*



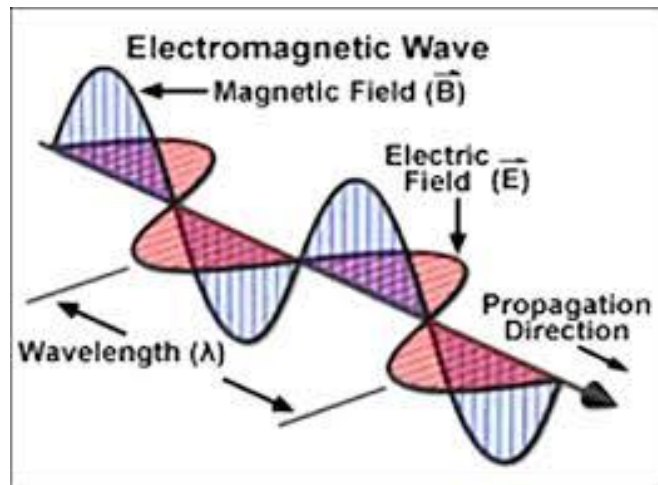
*Photo of Jupiter captured
by Voyager 1 (April, 1979)*

Remote sensing (Not only the earth)

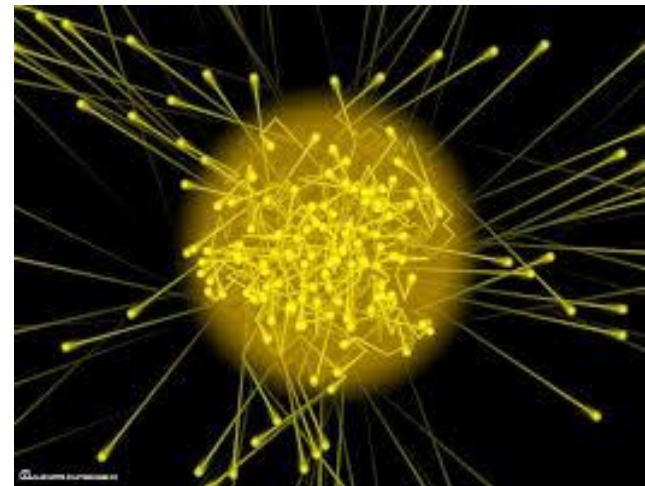
- Planetary geodesy (geology, topography..)
- Determine the distribution of several chemical elements

1.3 Physical basics of Remote sensing

The transfer of energy by electromagnetic radiation.



The wave mode



The particle mode (Photon)

The Wave Model of Electromagnetic Energy

- In the 1860s, James Clerk Maxwell (1831-1879) conceptualized electromagnetic radiation as an electromagnetic wave that travels through space at the speed of light, c , which is 3×10^8 meters per second (hereafter referred to as m s^{-1}) or 186,282.03 miles s^{-1} .
- The electromagnetic wave consists of two fluctuating fields, one **electric** and the other **magnetic**, hence the terminology electromagnetic. The two vectors are at right angles to one another, and both are perpendicular to the direction of travel.

Components of electromagnetic waves

- **Electromagnetic radiation** is generated when an electrical charge is accelerated.
- The **wavelength** of electromagnetic radiation depends upon the length of time that the charged particle is accelerated and its frequency depends on the number of accelerations per second.
- **Wavelength** is formally defined as the mean distance between maximums (or minimums) of a roughly periodic pattern and is normally measured in micrometers (μm) or nanometers (nm).
- **Frequency** is the number of wavelengths that pass a point per unit time. A wave that sends one crest by every second (completing one cycle) is said to have a frequency of one cycle per second or one hertz, abbreviated 1 Hz.

Wave Model of Electromagnetic Energy

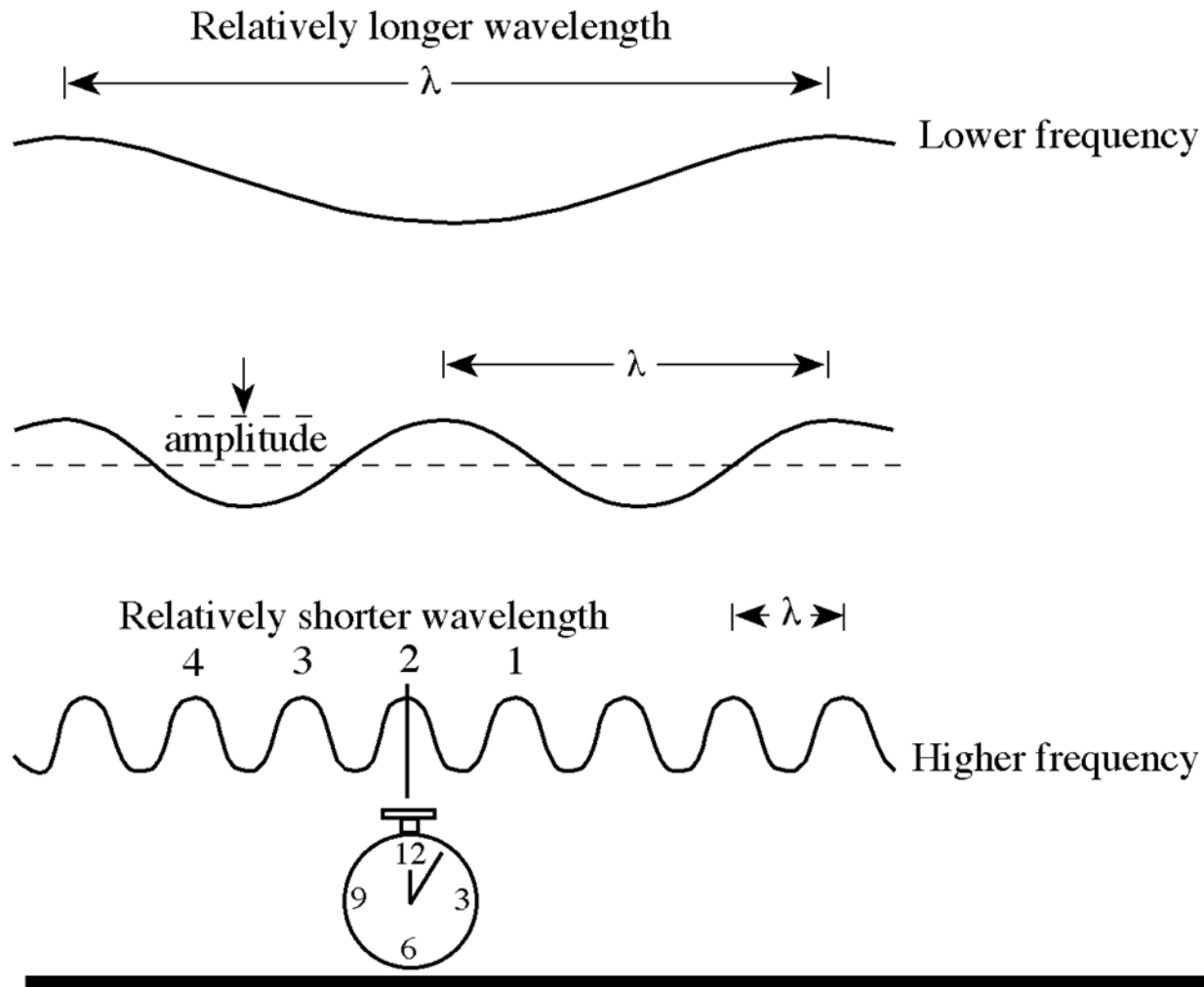
- The relationship between the wavelength, λ , and frequency, f , of electromagnetic radiation is based on the following formula, where c is the speed of light:

$$c = \lambda f$$

$$f = c / \lambda$$

- Note that frequency, f , is inversely proportional to wavelength, λ .
- The longer the wavelength, the lower the frequency, and vice-versa

Wave Model of Electromagnetic Energy



Stephen Boltzmann Law

- Using the wave model, it is possible to characterize the energy of the Sun which represents the initial source of most of the electromagnetic energy recorded by remote sensing systems (except radar). We may think of the Sun as a 6,000 °K blackbody (a theoretical construct which radiates energy at the maximum possible rate per unit area at each wavelength for any given temperature). The total emitted radiation (M_λ) from a blackbody is proportional to the fourth power of its absolute temperature. This is known as the Stefan-Boltzmann law and is expressed as:

$$M_\lambda = \sigma T^4$$

where σ is the Stefan-Boltzmann constant, $5.6697 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$. Thus, the amount of energy emitted by an object such as the Sun or the Earth is a function of its temperature.

Wien's Displacement Law

- In addition to computing the total amount of energy exiting a theoretical blackbody such as the Sun, we can also compute its dominant wavelength (λ_{max}) as:

$$\lambda_{max} = k / T$$

- where k is a constant equaling $2898 \mu\text{m } ^\circ\text{K}$, and T is temperature in degrees Kelvin.
- The Sun approximates a 6,000 k blackbody, therefore its dominant wavelength is:

$$0.483 \mu\text{m} = 2898 \mu\text{m } ^\circ\text{K} / 6000 ^\circ\text{K}$$

Particle Model of Electromagnetic Energy

- Niels Bohr (1885-1962) and Max Planck proposed the quantum theory of electromagnetic radiation.

$$Q = h c / \lambda = h f$$

Q ; the energy of a quantum (J)

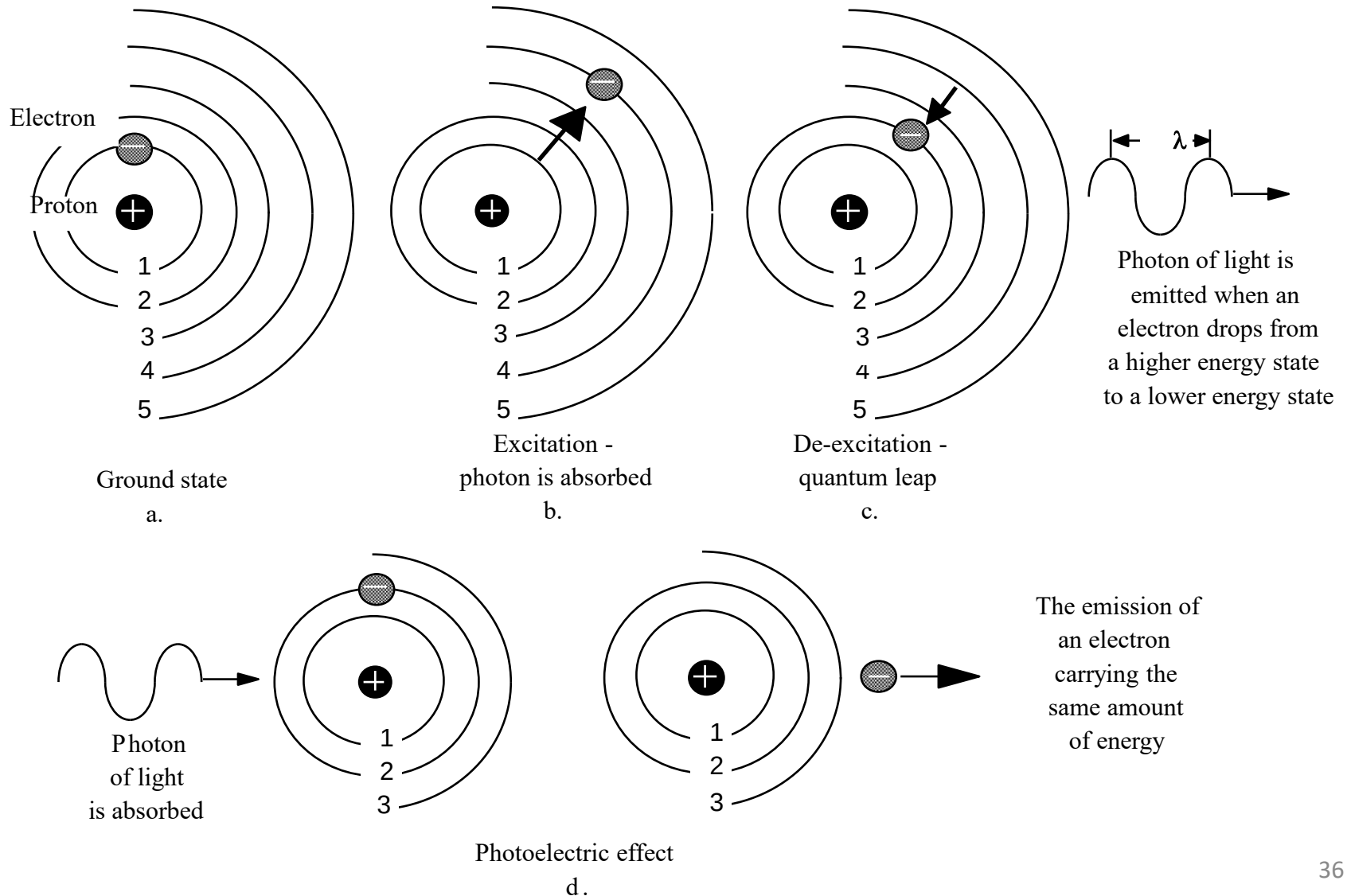
h ; the Planck constant = $6.626 \times 10^{-34} \text{ J s}^{-1}$

λ ; wave length.

- Thus, the energy of a quantum is inversely proportional to its wavelength, i.e. the longer the wavelength involved, the lower its energy content.

Mechanism Of Light Creation

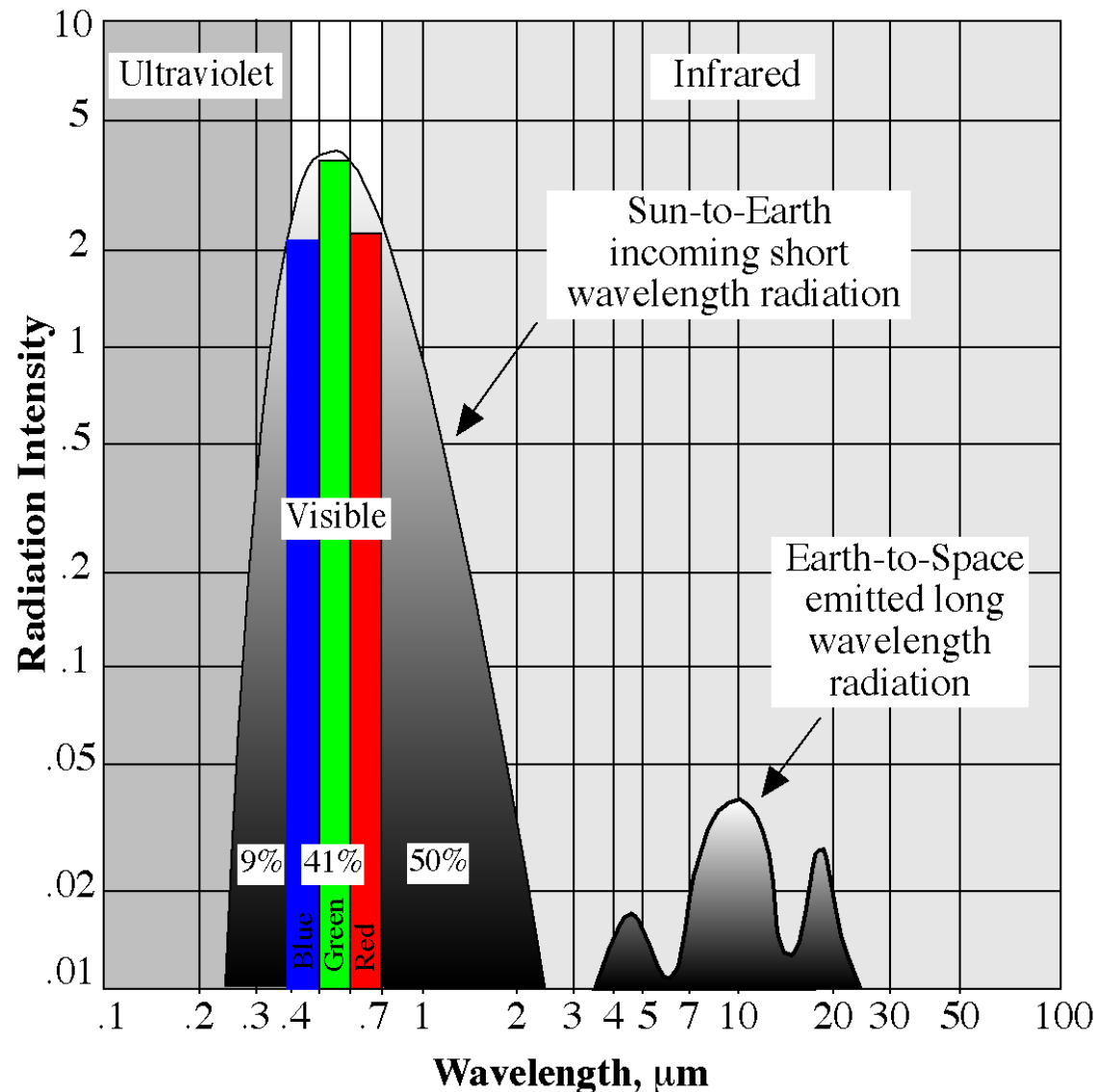
Creation of Light from Atomic Particles and the Photoelectric Effect



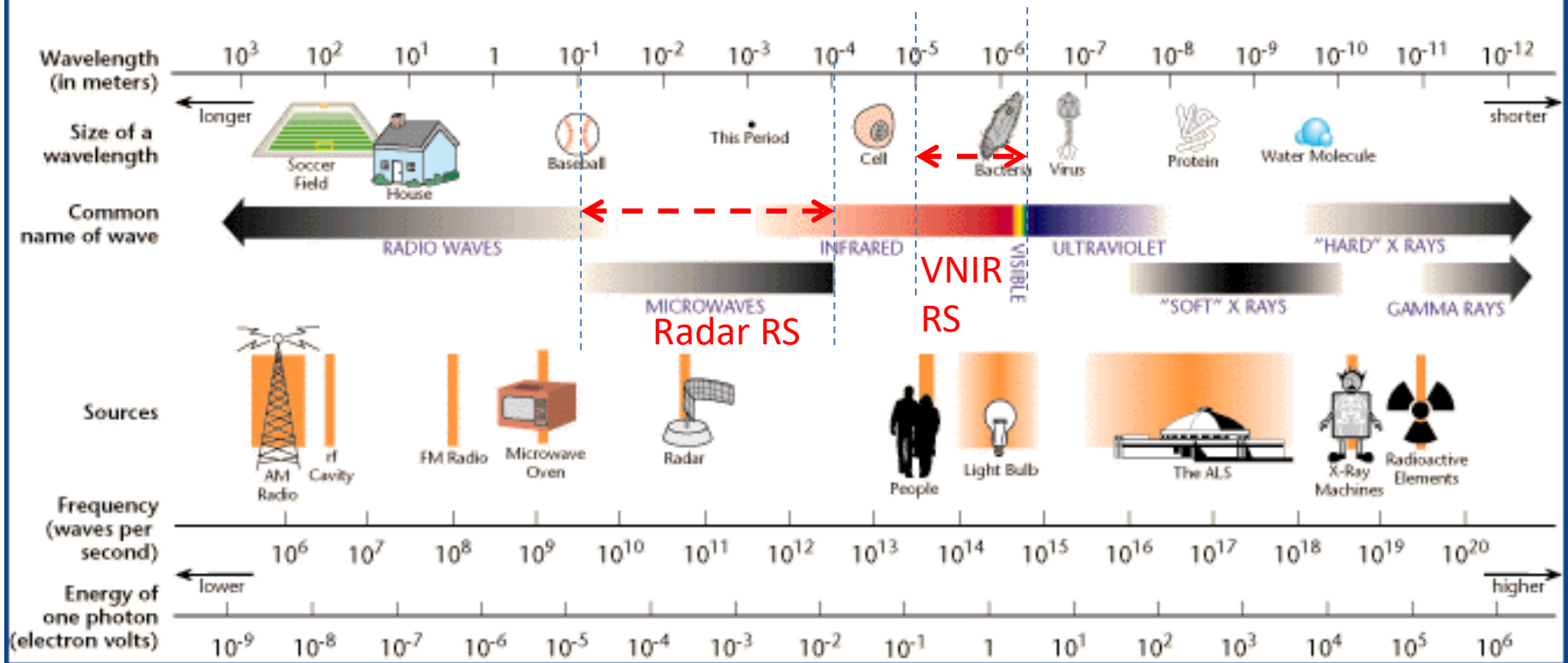
Wien's Displacement Law

The Sun: 6,000 K blackbody

- emits a dominant wavelength of $0.5\ \mu\text{m}$ (green light).
- produces 41% of its energy in the visible region from $0.4 - 0.7\ \mu\text{m}$ (blue, green, and red light).
- Produce 59% of the energy in wavelengths shorter than blue light ($<0.4\ \mu\text{m}$) and longer than red light ($>0.7\ \mu\text{m}$)
- Earth approximates a 300 K blackbody with a dominant wavelength of $9.7\ \mu\text{m}$.



THE ELECTROMAGNETIC SPECTRUM

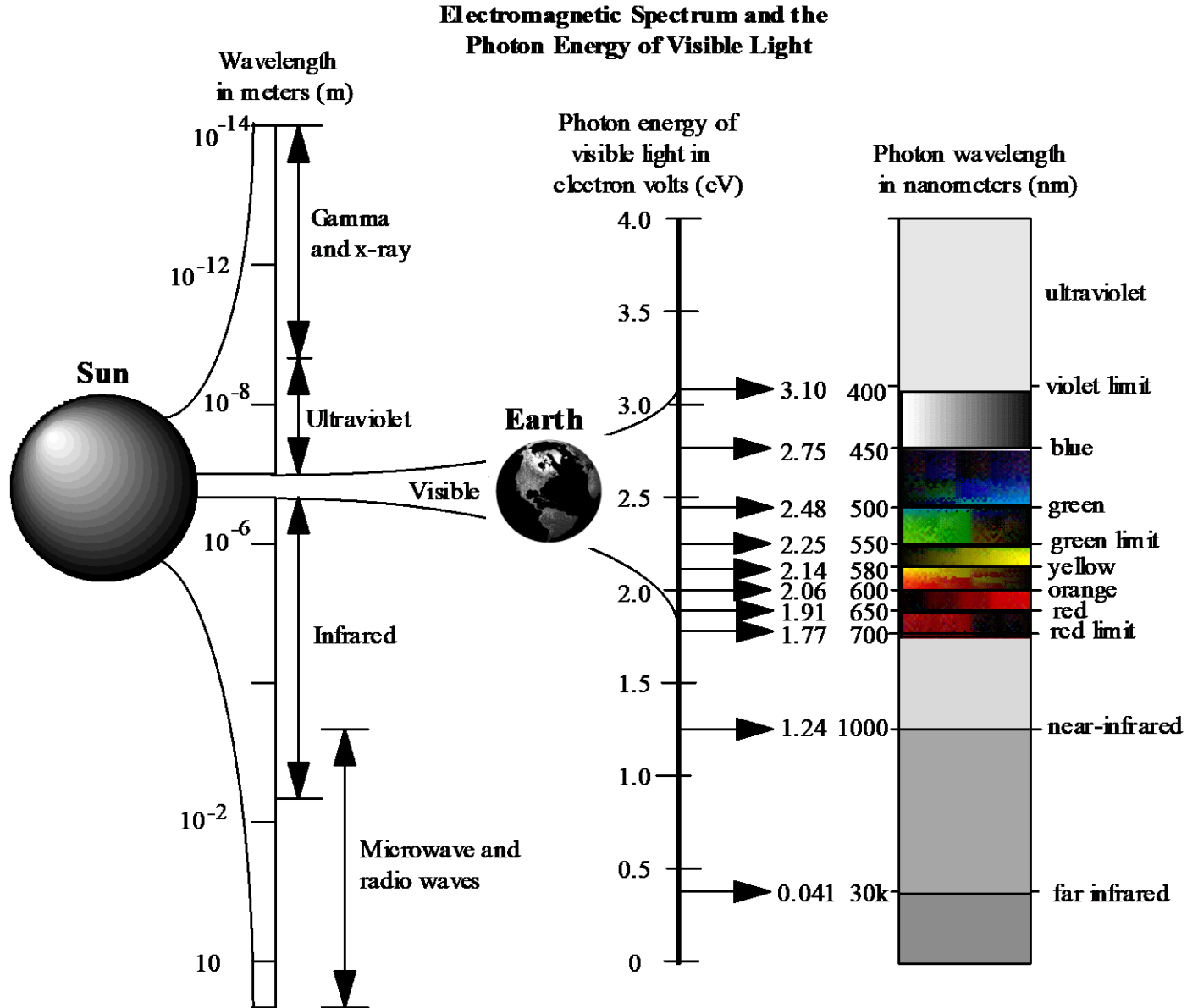


Energy radiated from sun (or active sensor)
 shorter wavelength (higher f) == higher energy
 longer wavelength (lower f) == lower energy

from http://rst.gsfc.nasa.gov/Intro/Part2_4.html

The Sun produces a continuous spectrum of energy from gamma rays to radio waves. The visible portion of the spectrum may be measured using wavelength (measured in nanometers or micrometers, i.e. nm or μm) or electron volts (eV) terminology. All units are interchangeable.

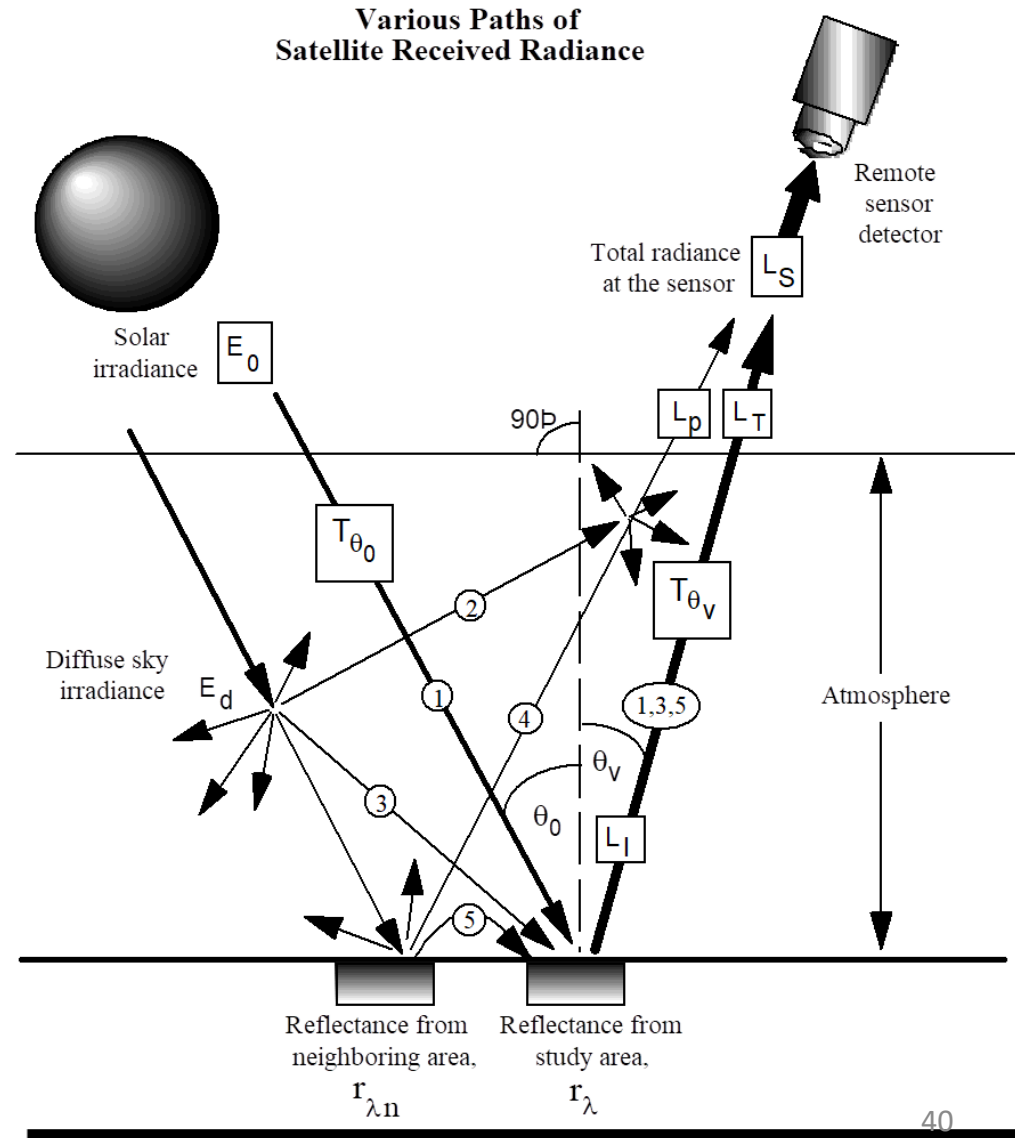
Electromagnetic Energy from The Sun



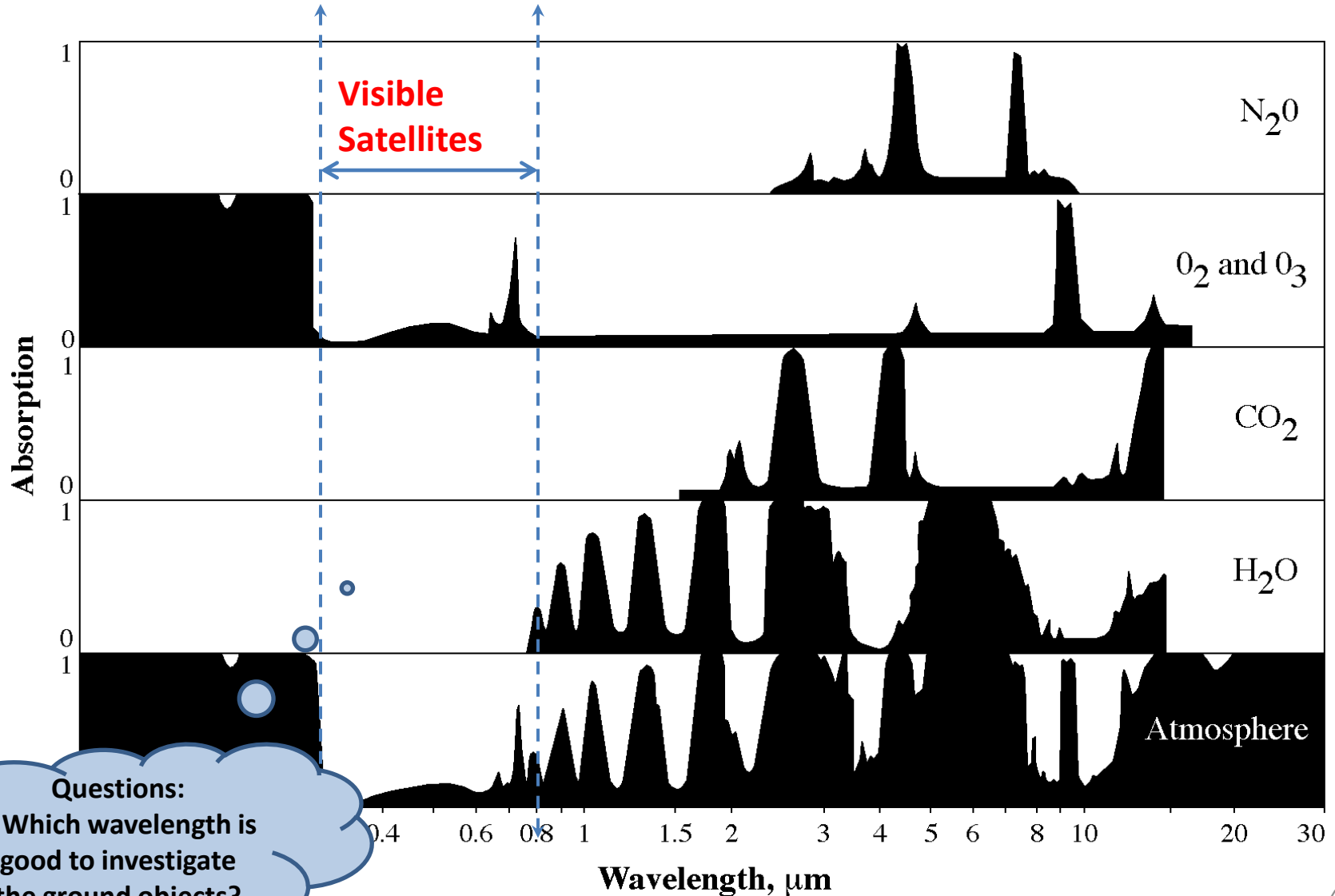
The Interaction Between Electromagnetic Wave And Atmosphere, Study Area, And Sensors

Electromagnetic Wave

- is radiated by atomic particles at the source (the Sun), Active radar
- propagates through the vacuum of space at the speed of light
- interacts with the Earth's atmosphere
- interacts with the Earth's surface
- interacts with the Earth's atmosphere once again
- finally reaches the remote sensor where it interacts with various optical systems, filters, emulsions, or detectors.



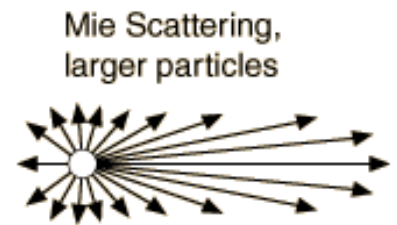
Absorption of the Sun's Incident Electromagnetic Energy in the Region from 0.1 to 30 μm by Various Atmospheric Gases



Scattering

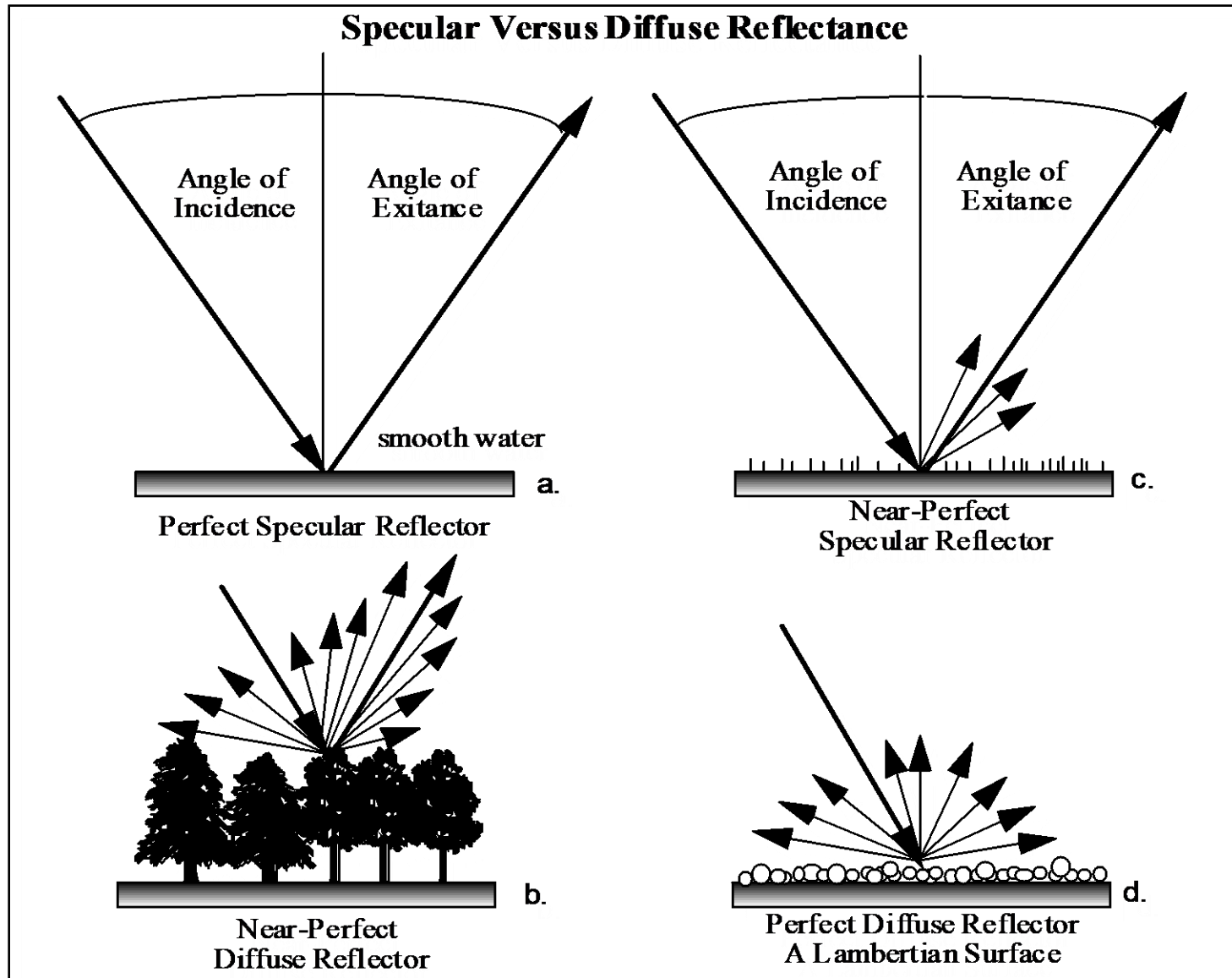
- Scatter differs from reflection in that the direction associated with scattering is unpredictable, whereas the direction of reflection is predictable. There are essentially three types of scattering:
 - Rayleigh, (particles smaller than EMR wavelengths)
Air molecules, N₂, O₂
 - Mie, (particles similar size as EMR wavelengths)
Dust
 - non-selective scattering.

Water in clouds

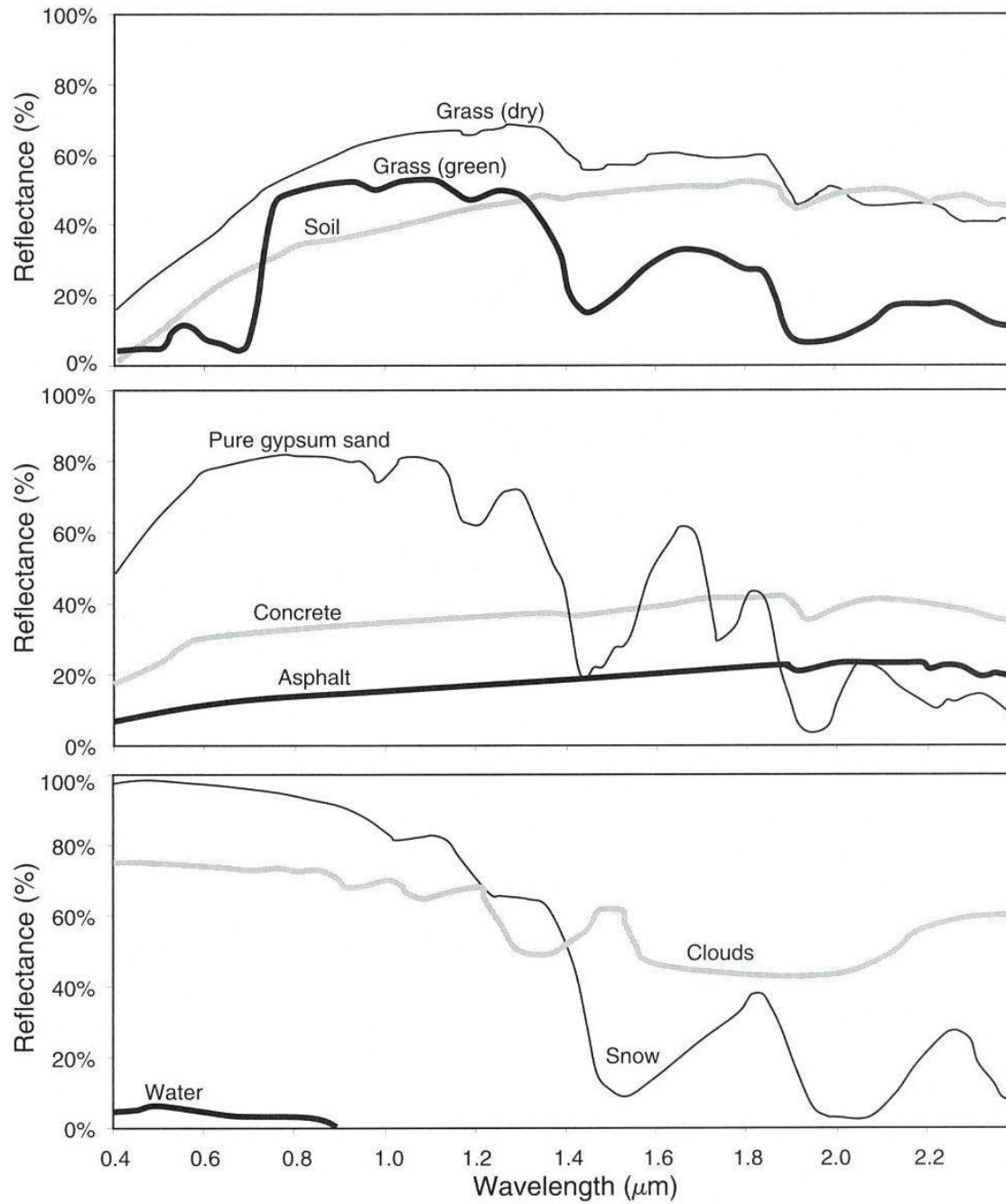


→ Direction of incident light

Reflectance



Spectral reflectance of surface components



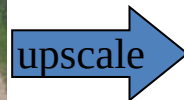
Source: Lillesand *et al.*, 2008

1.4 Remote sensing systems

- Based on the elevation of platforms



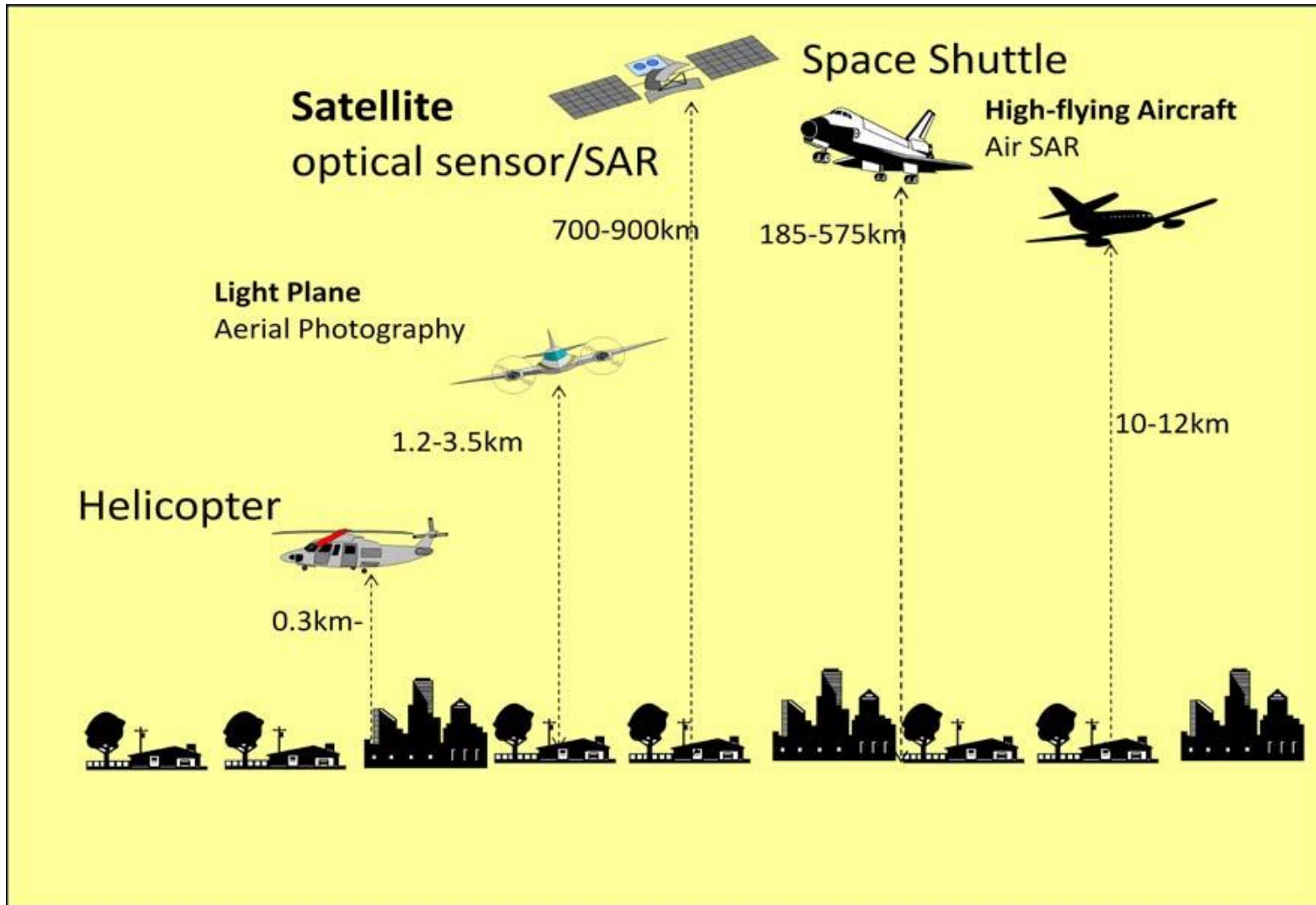
Ground remote sensing



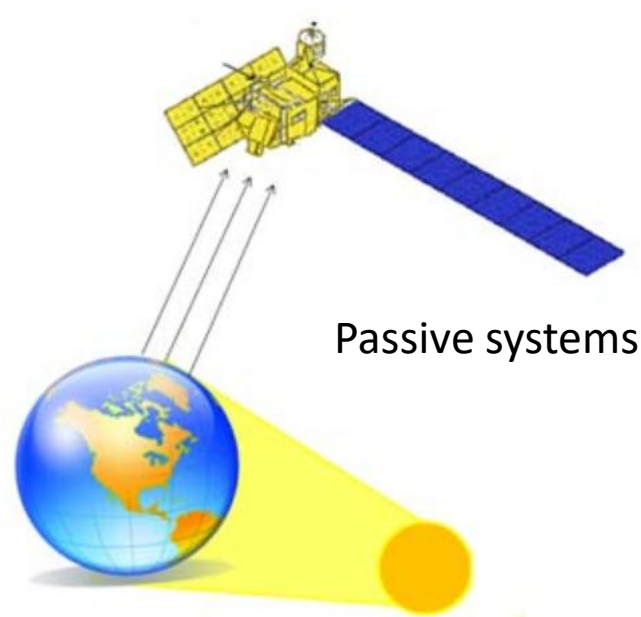
Aerial remote sensing (20 km)



***Space borne remote sensing
(> 200 km)***



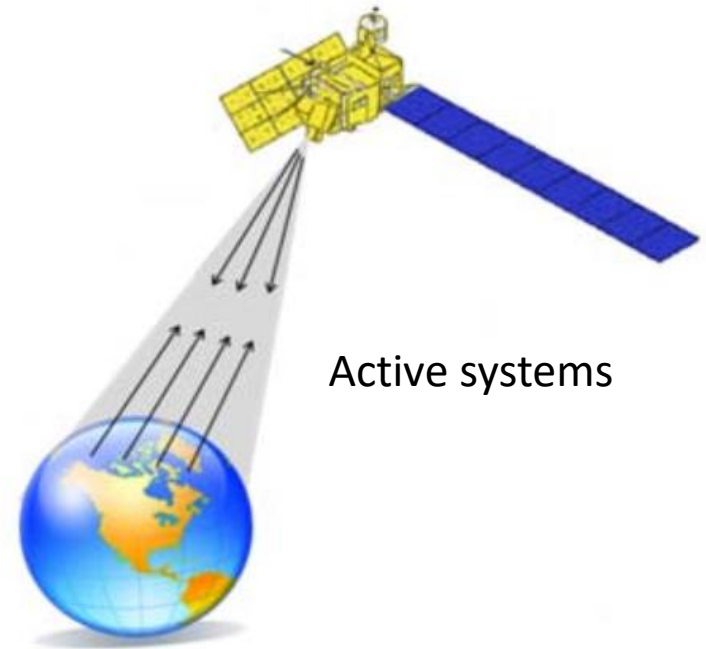
- Based on the sources of energy



Passive systems

Remote sensing systems which measure energy that is naturally available (sun) for reflection & emission.

LANDSAT, SPOT, SENTINEL 2,3, MODIS, NOAA



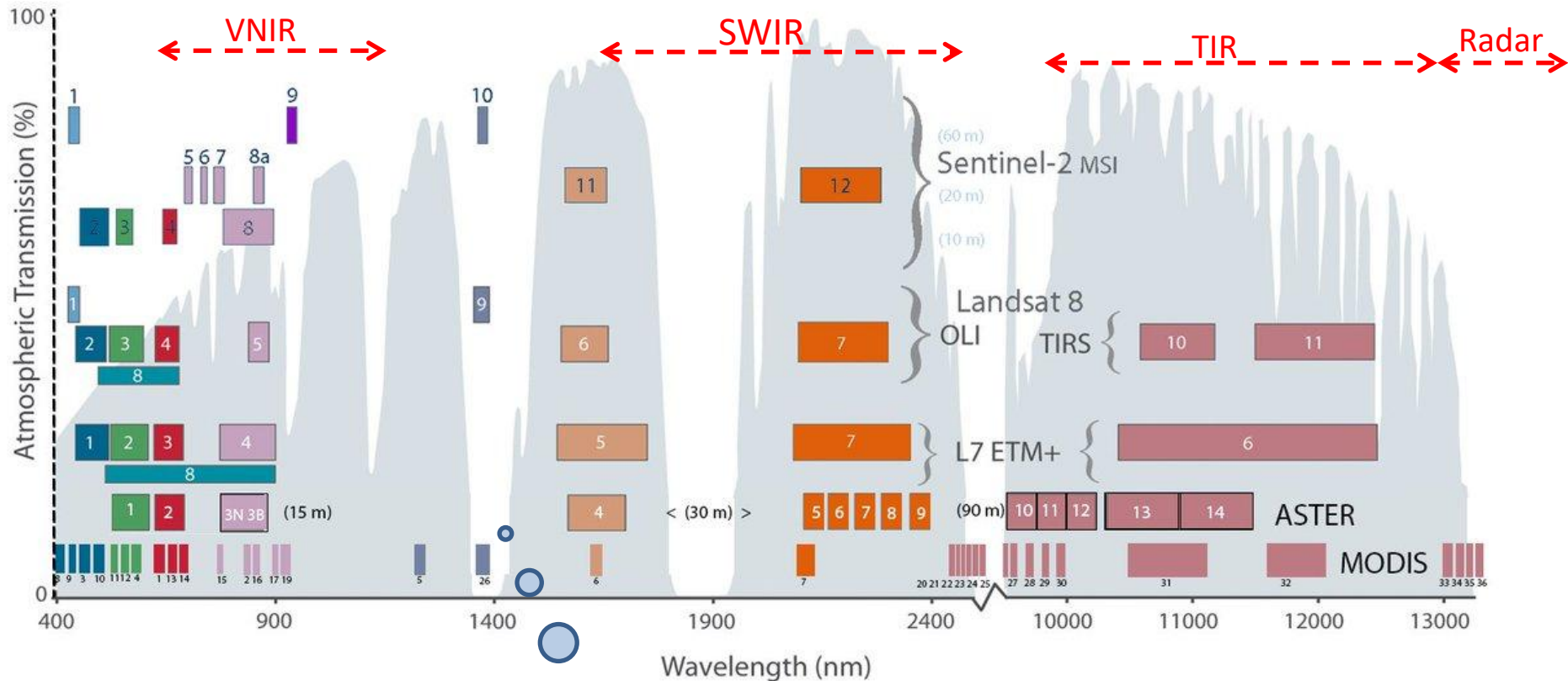
Active systems

Remote sensing systems provide their own energy source for illumination.

RADARSAT, ALOS PALSAR, SENTINEL 1..

- Based on the wavelength

Comparison of Landsat 7 and 8 bands with Sentinel-2



Why there are some bands here?

- **Based on the ground resolution**

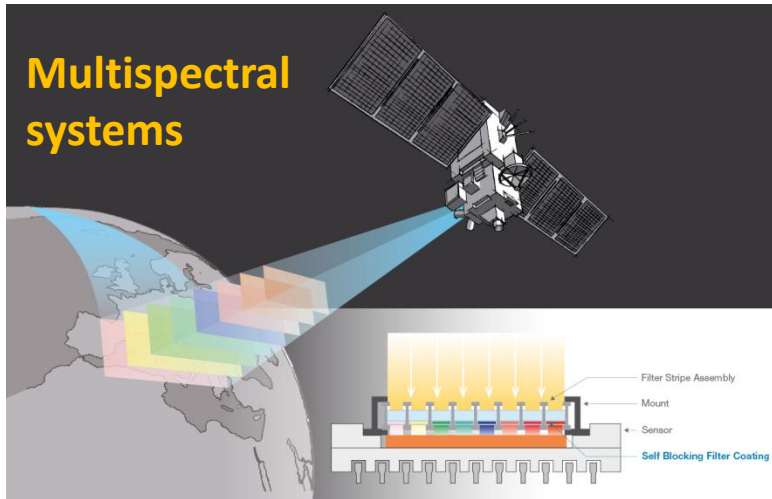
Very High Resolution				
Satellite (Sensor)	Spatial Resolution	Bands	Swath	Quantization
WorldView-2	0.46 m	1 (Panchromatic)	16.4 km (Nadir)	11 bit
	1.8 m	8 (Multispectral)	16.4 km (Nadir)	11 bit
WorldView-1	0.55 m	1 (Panchromatic)	17.6 km (Nadir)	11 bit
Quick Bird	0.61 m	1 (Panchromatic)	16.5 km (Nadir)	11 bit
	2.44 m	4 (B, G, R, NIR)	16.5 km (Nadir)	11 bit
IKONOS	0.82 m	1 (Panchromatic)	11.3 km (Nadir)	11 bit
	3.2 m	4 (B, G, R, NIR)	11.3 km (Nadir)	11 bit
FORMOSAT-2	2 m	1 (Panchromatic)	24 km	12 bit
	8 m	Multispectral	24 km	12 bit
ALOS (PRISM)	2.5 m	1 (Panchromatic)	70 km (Nadir)	8 bit
CARTOSAT-1	2.5 m	1 (Panchromatic)	55 km (Nadir)	10 bit
SPOT-5	2.5 m / 5 m	1 (Panchromatic)	60 km-80 km (Nadir)	8 bit
Planet	0.5 m/ 5m	5 (Multispectral)	5 km – 77 km	12 bit

High Resolution				
Satellite (Sensor)	Spatial Resolution	Bands	Swath	Quantization
ASTER (VNIR)	15 m	3 (G, R, NIR)	60 km	8 bit
ASTER (SWIR)	30 m	6	60 km	8 bit
ASTER (TIR)	90 m	5	60 km	8 bit
Landsat ETM+	30 m	7	185 km	8 bit
Medium Resolution				
Satellite (Sensor)	Spatial Resolution	Bands	Swath	Quantization
MODIS	250 m	(1-2)	2330 km (cross track) by 10 km (along track and nadir)	12 bit
	500 m	(3-7)		
	1000 m	(8-36)		

Low Resolution

Himawari 8,9, NOAA....> 1000 m

- Based on the number of bands



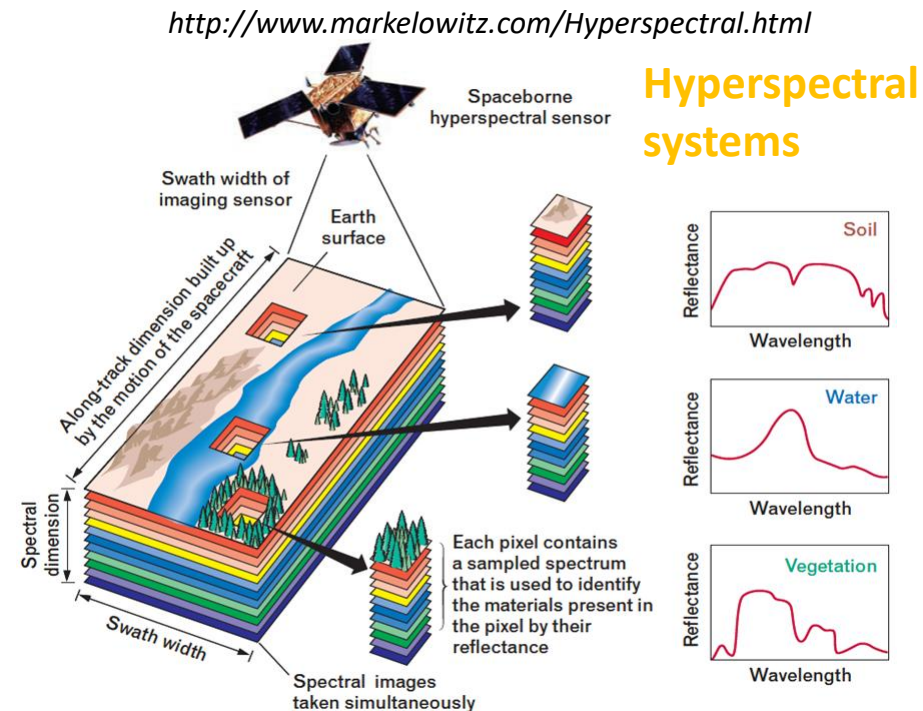
<https://www.opticsbalzers.com/en/applications/remote-multi-spectral-imaging.html>

Systems acquire multispectral image consists of several spectral bands of data (Lansat, Spot, Alos, Sentinel, Modis)



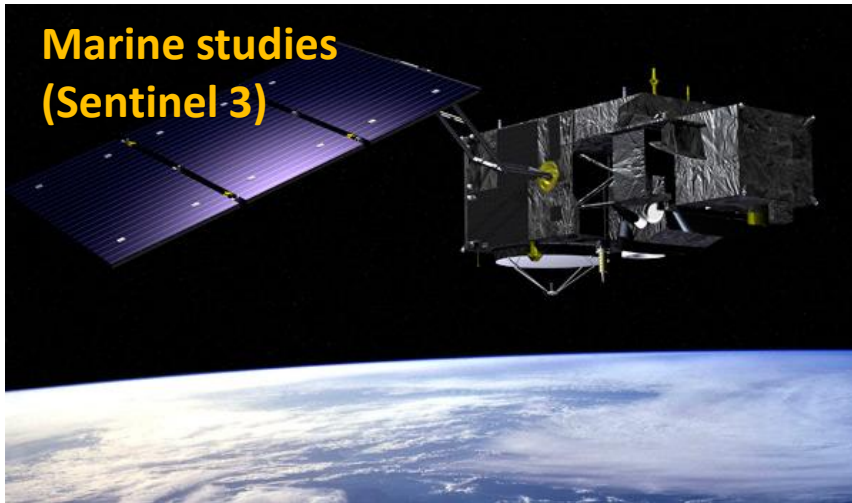
Panchromatic systems

The total light energy in the visible spectrum are recorded in one band (IKONOS PAN, SPOT HRV-PAN ALOS-PRISM)

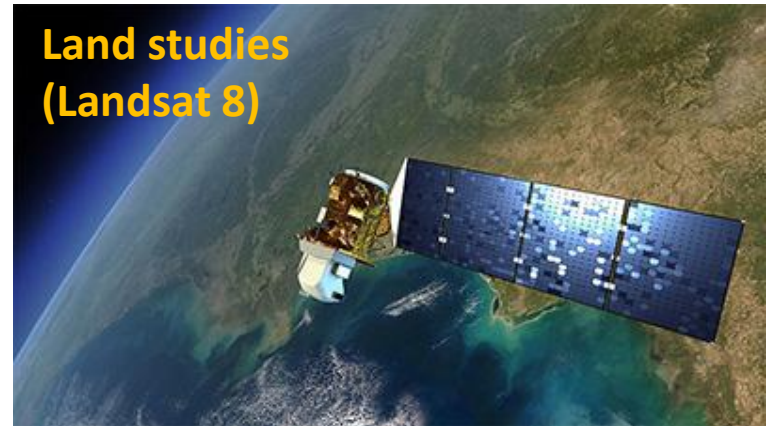


Detect hundred of narrow, contiguous spectral bands simultaneously: *AIS*, 128 bands, 10 nm, 8 m; *AVIRIS* 224 bands, *MightySat II* 256 bands, *PROBA*

- **Based on the application**



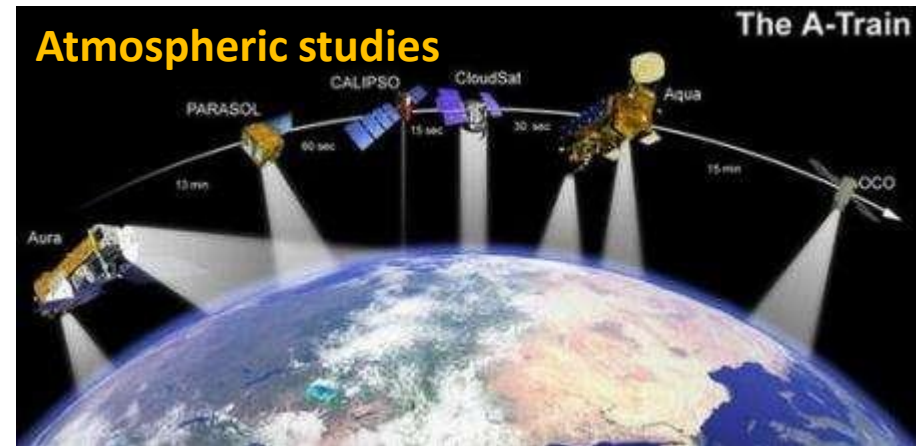
Designed to study Ocean (SENTINEL 3, MERIS, GEOS, NOAA, TRMM)



Designed to separate 3 main components of the Earth surface (SENTINEL 1,2, LANDSAT, SPOT, QUICKBIRD...)



Designed to study Ocean (HIMAWARI, METOP, NOAA, TRMM)

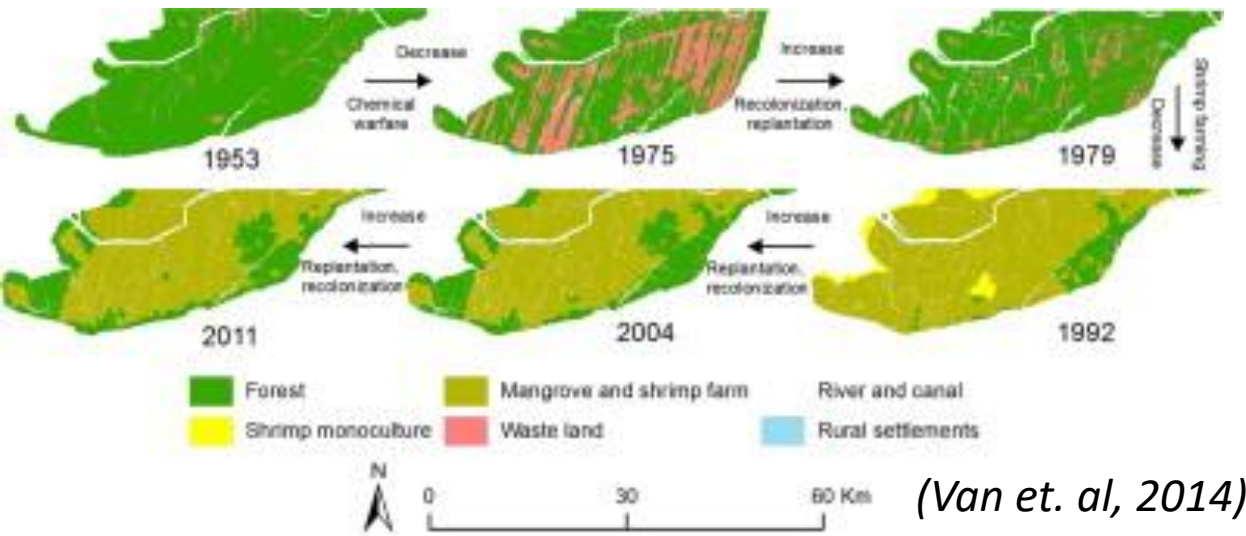


Designed to investigate the nature of atmosphere

1.5 Application of Remote Sensing

- Environmental: climate, ecosystem & habitat mapping, land cover change, hazard mapping and monitoring, vegetation, carbon cycle, oceans, ice
- Commercial: telecomms, agriculture, geology and petroleum, mapping
- Military: reconnaissance, mapping, navigation (GPS)
- Weather monitoring and prediction.....

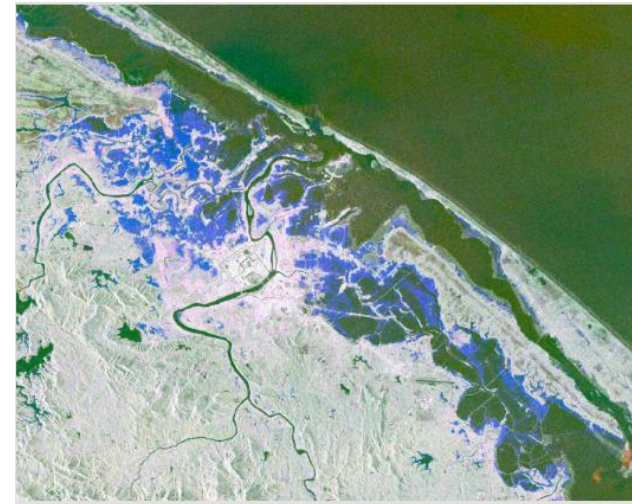
Forest change using Landsat



(Van et. al, 2014)

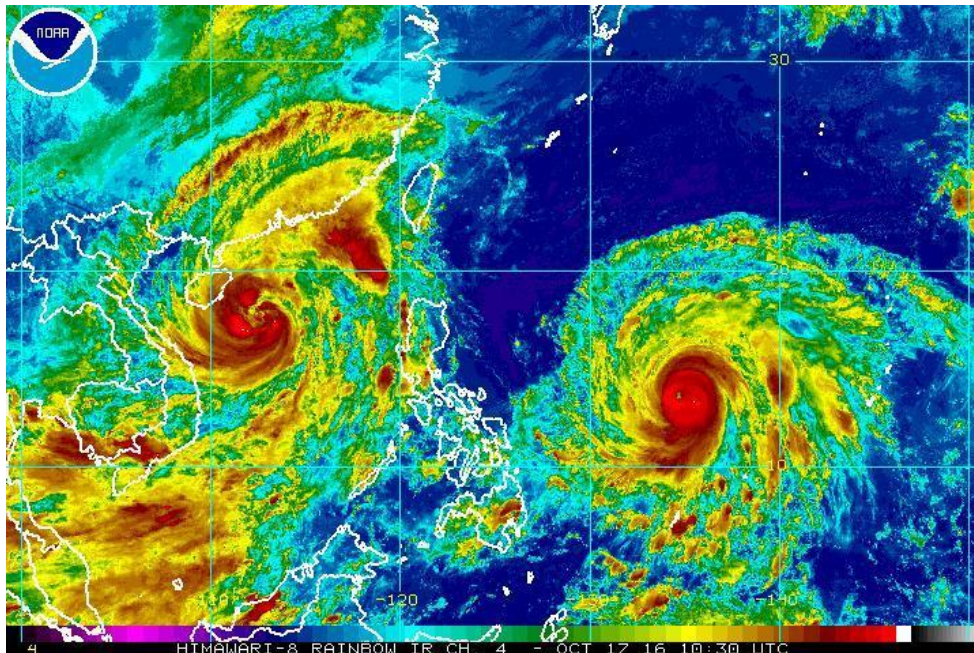
Changes in mangrove vegetation area and character in a war and land use change affected region of Vietnam (Mui Ca Mau) over six decades

Local, regional scale

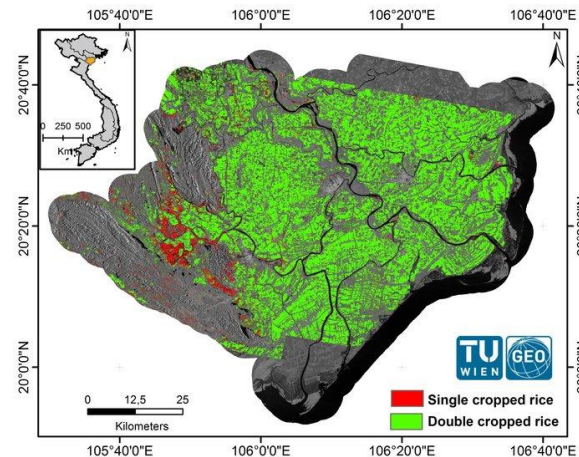


Acquired Nov. 11, 2017

Flooding in Hue, 2017 using Sentinel 1
<https://disasters.nasa.gov/typhoon-damrey-2017/esa-copernicus-sentinel-1-sar-used-flood-detection-maps>



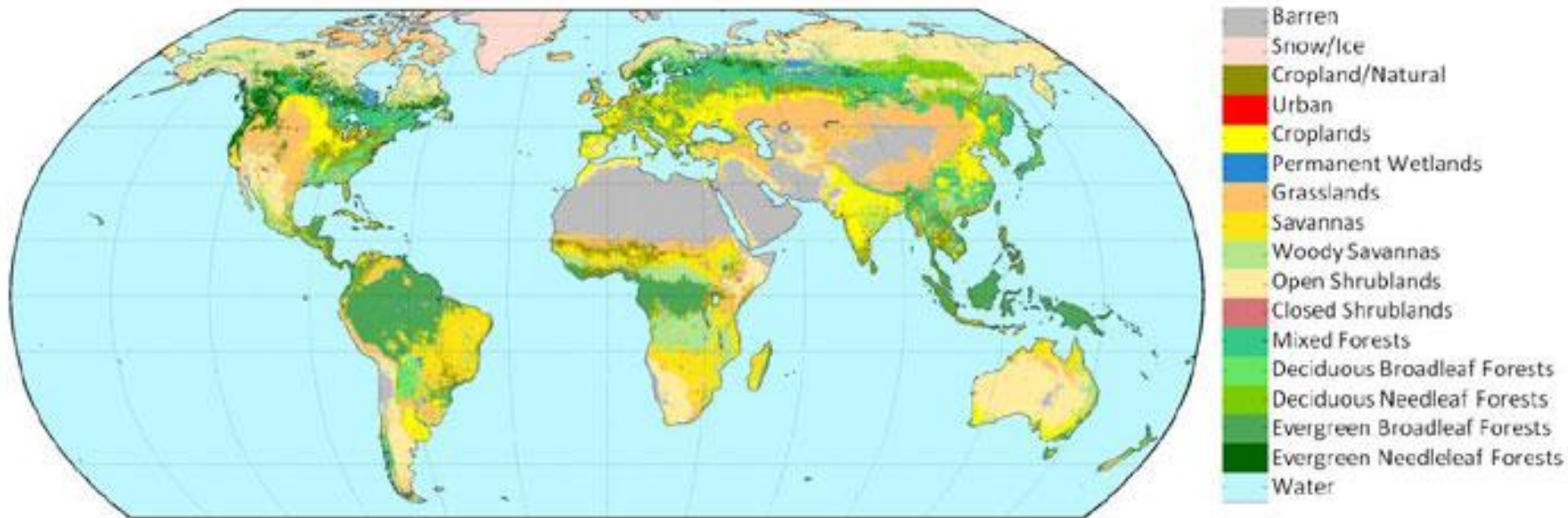
Typhoon
Sarika and
Haima
17/10/2017
using
Himawari 8
([Andrew V. Pestano](#),
2017)



Rice map in Red river delta using
Sentinel-1 data in 2015

(<https://www.esa.int>)

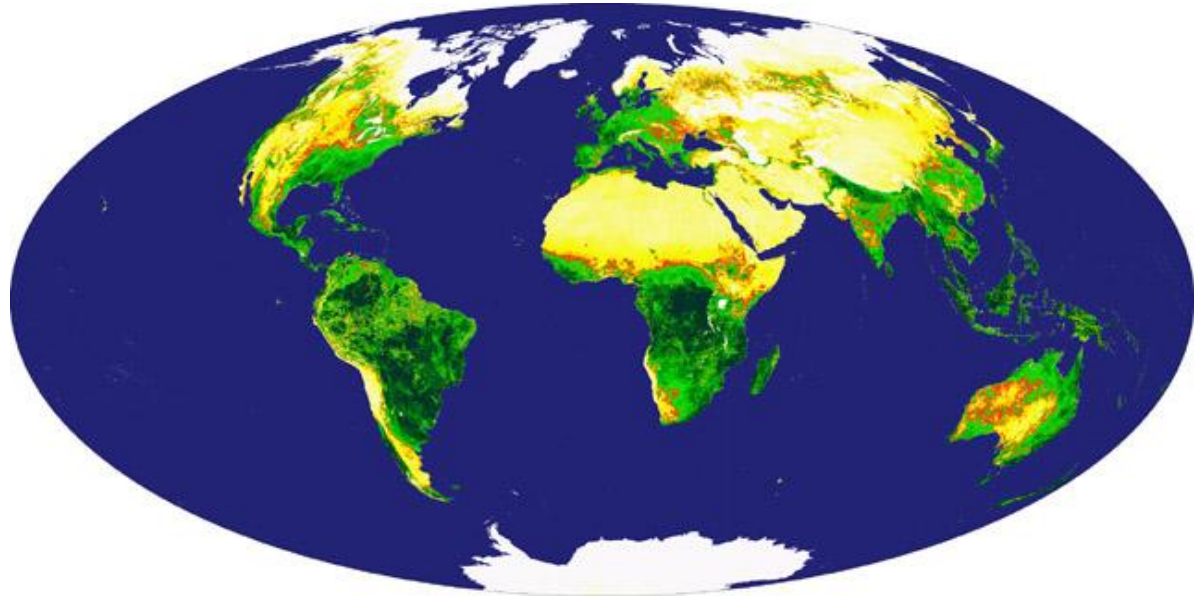
Global Land Cover



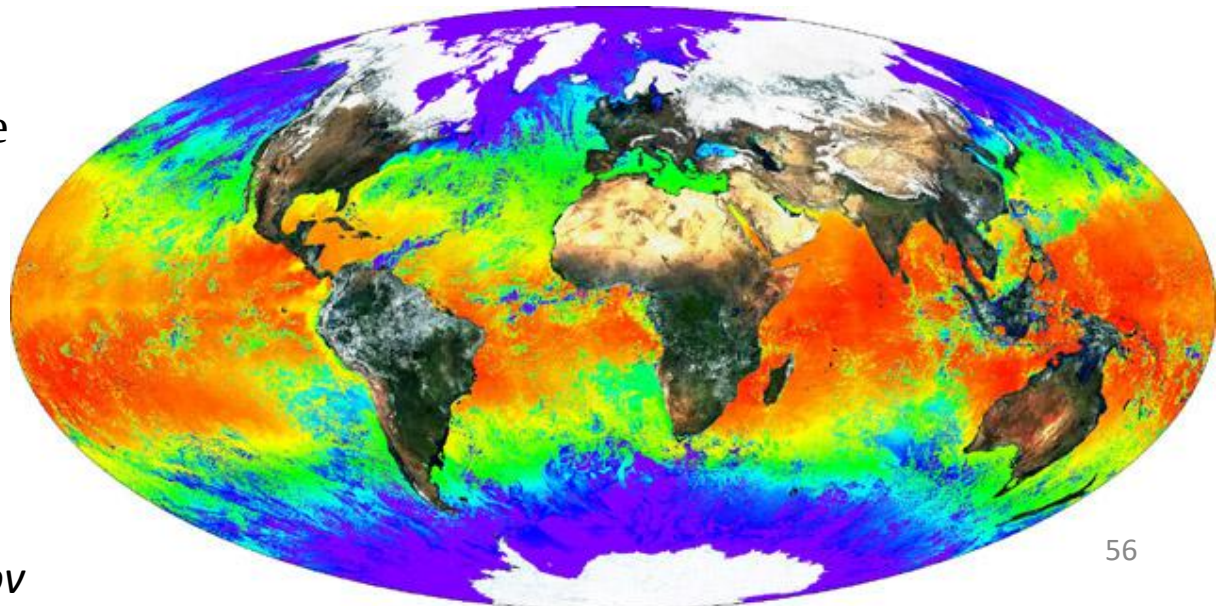
•Global land cover from MODIS images

([Broxton](#) and [Xubin Zeng](#), 2014)

- Global maps of vegetation amount from MODIS instrument



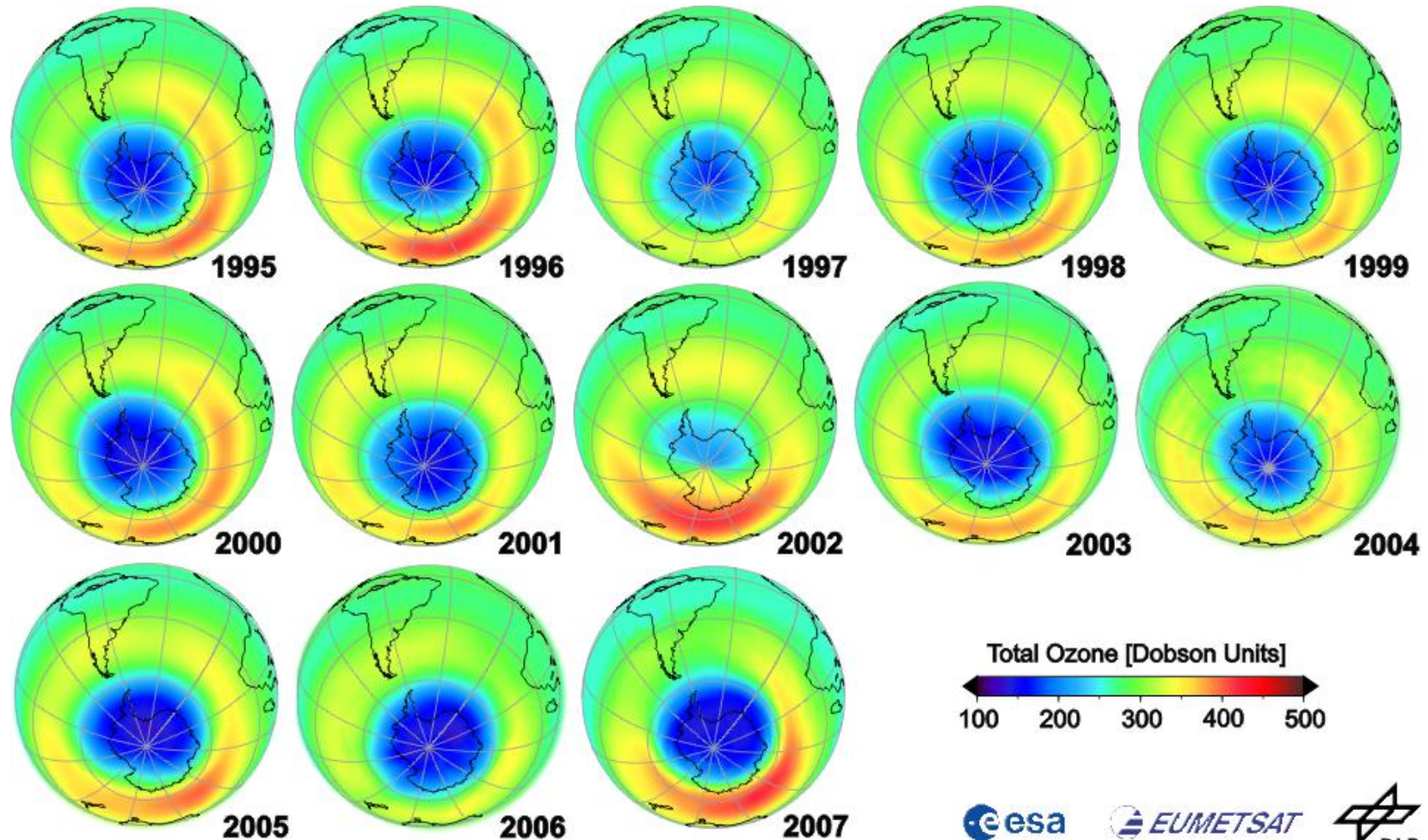
- Global maps of sea surface temperature from MODIS instrument



Atmospheric study

Monitoring the Antarctic Ozone Hole by GOME, SCIAMACHY and GOME-2

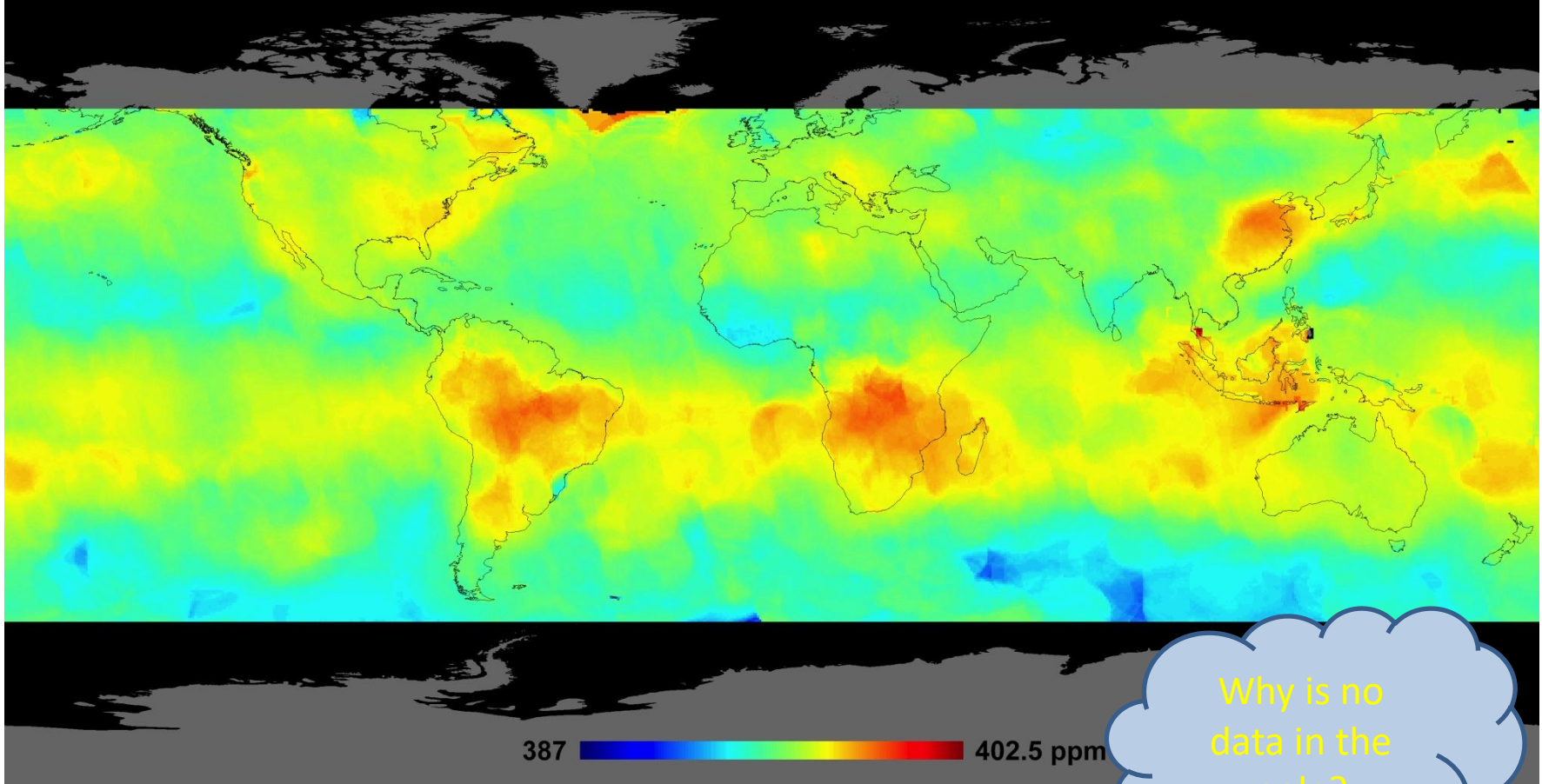
Total Ozone Monthly Mean, September 1995 - 2007



(Joseph D'Aleo, CCM, AMS Fellow⁵⁷)

Atmospheric study

Averaged Carbon Dioxide Concentration Oct 1 - Nov 11, 2014 from OCO-2



Why is no
data in the
pole?

Remarks

- Remote sensing system: Sources, Lights, Targets, Sensors
- Interaction of lights: With particles in atmosphere, with targets, with sensors

Thank you for your attention!